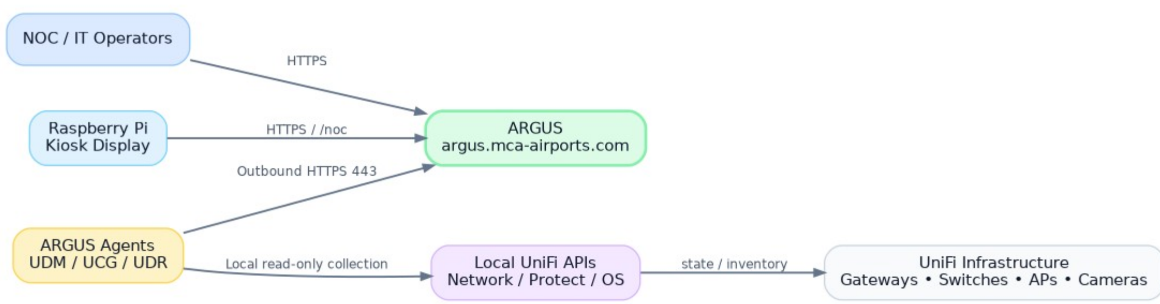
ARGUS

UniFi Infrastructure Watchtower

Architecture & Functional Specification v1.0

Monitoring, health, VPN/WAN observability, configuration summaries, incident management and NOC wallboard



Conceptual system context

Target platform: Ubuntu Server 26.04 on AWS | Public URL: argus.mca-airports.com

Primary visual constraint: NOC/Kiosk views must fit comfortably at 1920×1080 or lower without scrolling.

Document Control

Field	Value
Document	ARGUS Architecture & Functional Specification
Version	1.0 - Initial architecture baseline
Date	August 12, 2026
Status	Working specification / implementation baseline
Target scale	Approximately 100 UniFi installations at initial rollout
Server	AWS EC2 / Ubuntu Server 26.04
Primary URL	argus.mca-airports.com
Client delivery	MCA-Secure-Downloads-install
Design posture	Read-only observability of UniFi; outbound-only agents

Core design statement
ARGUS observes UniFi. It does not replace Site Manager and it does not make configuration changes to Network or Protect in V1. The system is designed to answer, quickly and visually: Where is the problem? What is failing? Since when? What is affected?

Contents

1. Executive Summary

ARGUS is a centralized observability platform for a distributed UniFi estate of roughly 100 installations. A lightweight Agent installed on UDM, UCG and UDR-class consoles performs local read-only inspection of UniFi Network, Protect and console health, then reports normalized state to a public ARGUS server in AWS over outbound HTTPS only.

The platform is deliberately optimized for operational clarity rather than deep performance analytics. Its primary job is to identify unhealthy sites, WANs, VPN tunnels, gateways, switches, access points, cameras and agents; correlate failures into incidents; preserve searchable history; and provide an extremely clear NOC wallboard that remains usable on a Raspberry Pi kiosk at 1920×1080 or lower.

- Agent heartbeat and dead-man monitoring.
- Automatic discovery followed by explicit console approval and device selection.
- WAN and VPN health with both control-plane status and data-plane probes.
- Current-state storage plus state-change/event history instead of storing every repetitive sample.
- Site/network configuration summary with secret redaction and change hashing.
- Incident grouping, acknowledgements, maintenance windows, event logs and immutable audit logs.
- Separate administrative UI and NOC/Kiosk UI.
- One-time enrollment tokens and unique Agent credentials; future mTLS capability.

2. Scope and Design Principles

2.1 V1 scope

- UniFi console enrollment and approval.
- Network / Protect inventory discovery.
- Selective monitoring by device and monitoring profile.
- Console, Agent, WAN, VPN, Network, Protect and device health.
- VPN data-plane probe targets.
- Network configuration summaries: networks/VLANs, WAN metadata, WiFi SSIDs, VPN metadata and high-level firewall-zone counts when available.
- Event, alert, incident and audit models.
- Searchable/filterable logs.
- Light, Dark and System themes.
- NOC route optimized for 1080p kiosk use.
- Email notifications; architecture prepared for Teams/webhooks later.

2.2 Explicit non-goals for V1

- No configuration writes to UniFi Network/Protect.
- No storage of WiFi passwords, VPN PSKs, private keys, RADIUS secrets or administrative passwords.
- No attempt to reproduce every UniFi screen or setting.
- No full L2 topology/root-cause graph in V1.
- No collection of high-volume client traffic analytics unless later required for a specific health rule.

2.3 Design principles

Principle	Implementation consequence
Operational clarity first	Dashboard answers where/what/since-when before exposing detail.

Outbound-only sites	Agents initiate HTTPS; no inbound ports required at customer/site networks.
Read-only UniFi posture	Collector permissions are scoped to monitoring where possible.
State changes over raw samples	Frequent checks update current state; history records meaningful transitions.
Unknown is not Offline	If an Agent disappears, dependent devices become UNKNOWN unless independent evidence proves outage.
Kiosk is a first-class client	/noc is separately designed for 1080p, distance viewing and no scroll.
Scale by profiles	~100 sites inherit policies rather than being individually hand-tuned.
Security by per-agent identity	Compromise/revocation of one site does not rotate credentials across the estate.

3. Functional Architecture

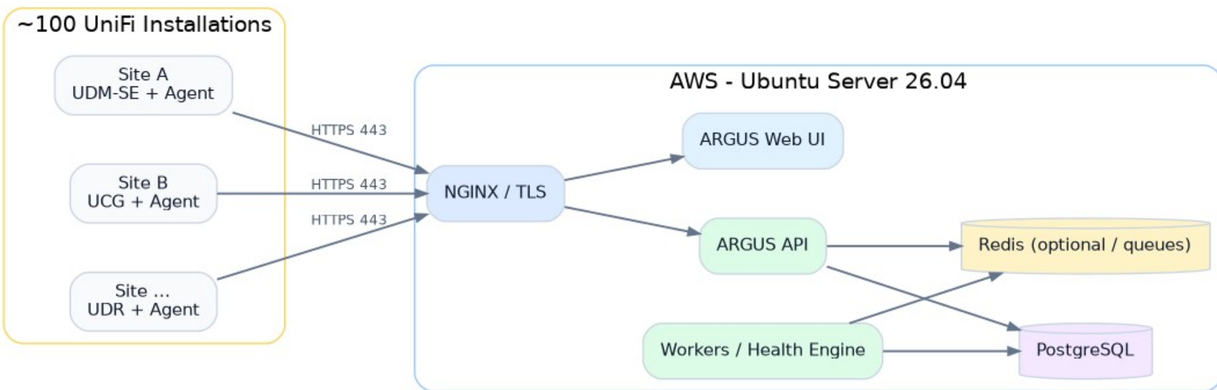


Figure 3.1 - AWS and distributed-site deployment topology

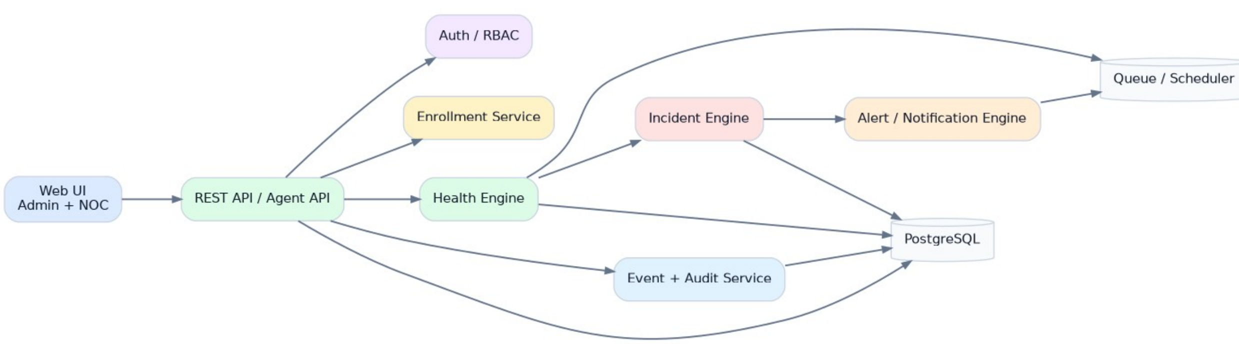


Figure 3.2 - ARGUS server component architecture

The initial deployment may use Docker Compose on a single EC2 instance while preserving service boundaries that can later be separated. PostgreSQL is mandatory. Redis is recommended for queues/scheduling once asynchronous alerting, notification fan-out and command dispatch become active; it can be omitted during the earliest prototype if simplicity is preferred.

3.1 Recommended server components

Component	Responsibility
NGINX	TLS termination, security headers, reverse proxy, static UI delivery.
argus-api	REST endpoints for agents and administrative clients.
argus-web	React/TypeScript administrative and NOC interface.
argus-worker	Scheduled health evaluation, incident correlation, notifications, retention jobs.
PostgreSQL	Persistent relational state, history, incidents, audit and configuration summaries.
Redis	Queue/cache/short-lived coordination when enabled.

4. Hierarchy and Domain Model

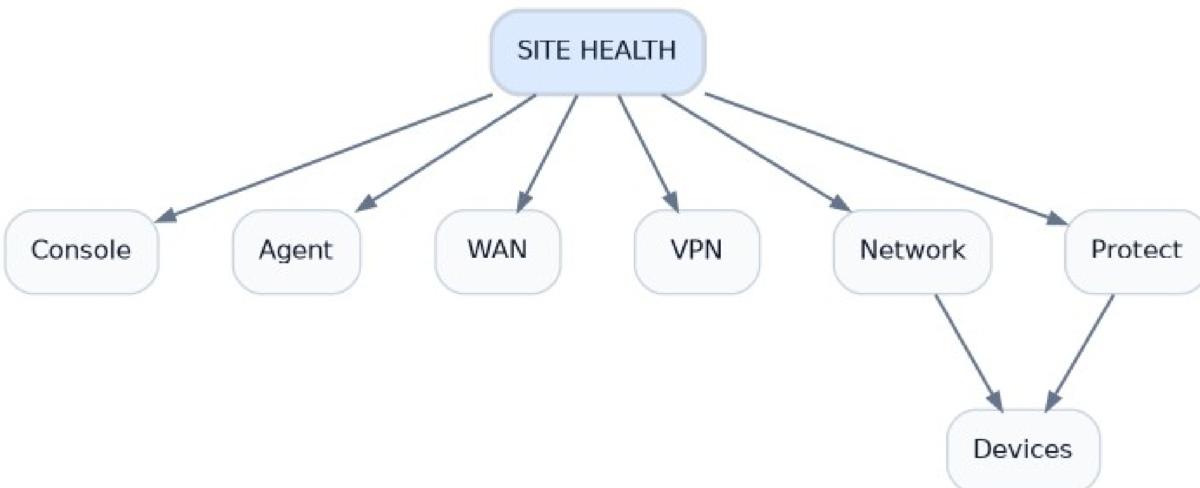


Figure 4.1 - Site health hierarchy

ARGUS should use Site as its primary operational unit. A Site can correspond to an airport terminal, store group, office or other logical installation. A Site normally contains one managed UniFi Console in V1, but the data model must permit multiple consoles if later needed.

Entity	Purpose
Organization	Top-level tenant boundary; initially Master ConcessionAir.
Site	Operational location shown in NOC and health rollups.
Console	UDM/UCG/UDR/other UniFi control-plane host.
Agent	ARGUS software identity associated with a console.
Application	Network, Protect or other locally available UniFi application.
Device	Gateway, switch, AP, camera and selected future UniFi asset.
WAN interface	Internet uplink monitored independently.
VPN tunnel	First-class monitored object with control/data-plane health.
Monitoring profile	Inherited cadence, thresholds and default device-selection rules.

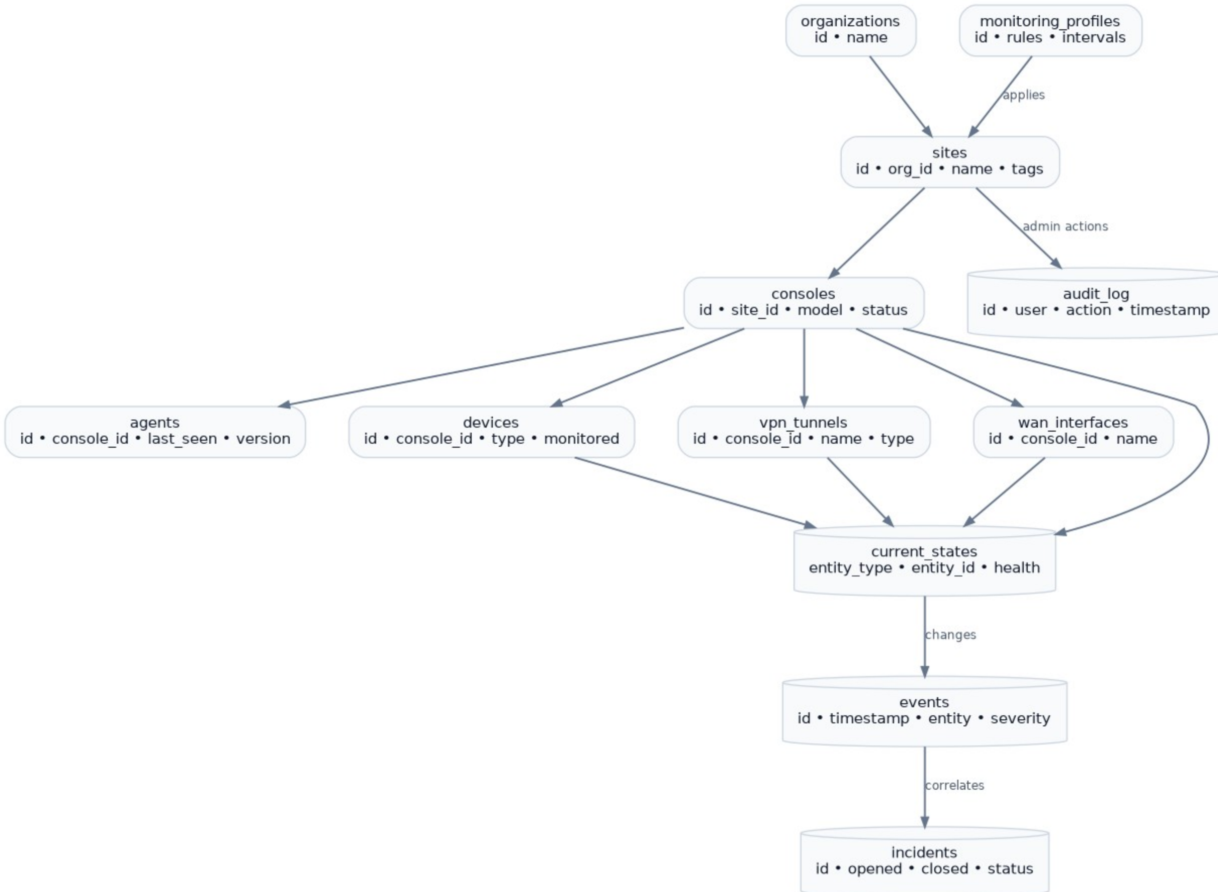


Figure 4.2 - Core relational data model

5. Agent Architecture

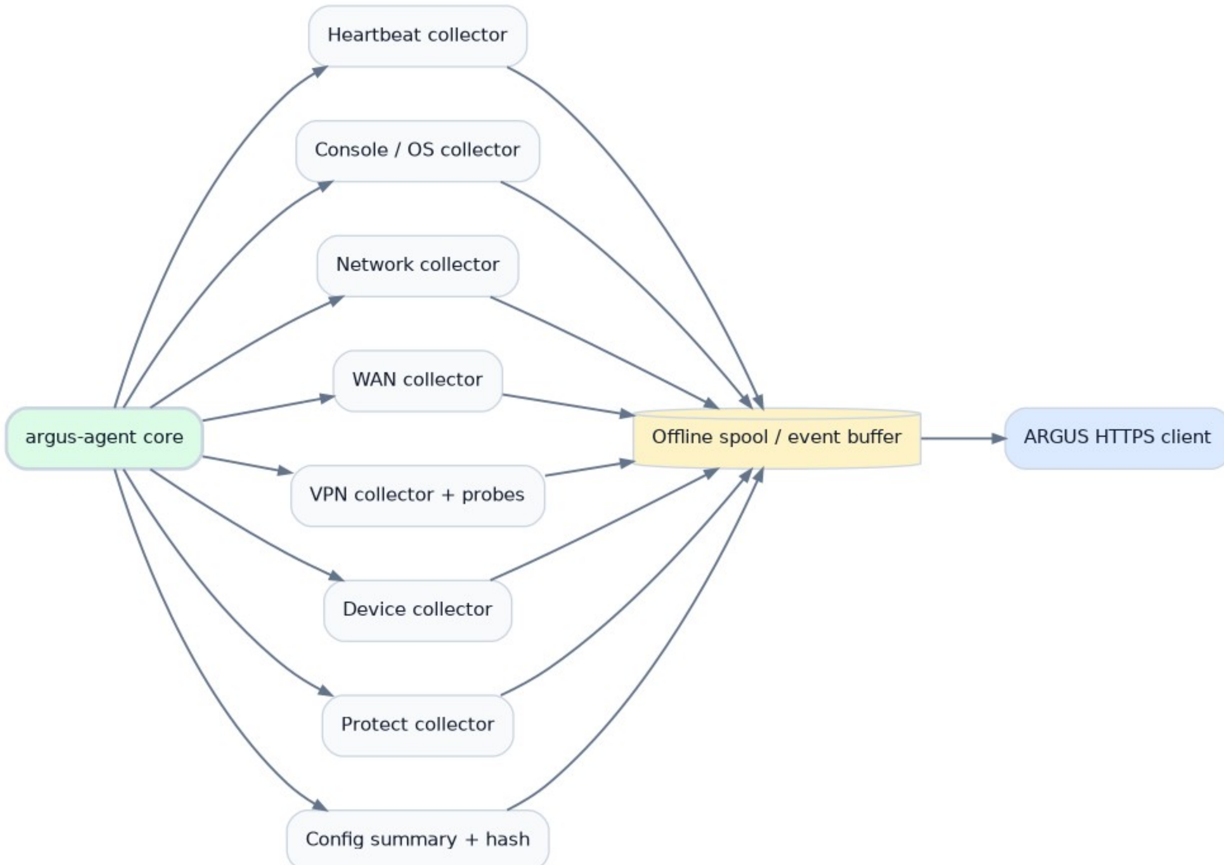


Figure 5.1 - Agent internal components

The Agent should be a small, self-supervising service with a local spool. Python is acceptable for V1 because it accelerates API integration and iteration; a later Go implementation remains possible if a single static binary becomes operationally advantageous.

Collector	Typical cadence	Output
Heartbeat	15 s + jitter	Agent version, timestamp, console identity, basic availability.
Console / OS	30-60 s	Console model/version/state and resource-health indicators needed for monitoring.
WAN	15-30 s	Interface state and approved latency/loss probes.
VPN	15-30 s	Tunnel state plus optional remote probe results.
Devices	30-60 s	Selected-device health and reachability/state.
Protect	30-60 s	Selected camera/device health and recording-relevant state where available.
Inventory	~5 min incremental / ~1 h full	Discovered devices, applications and metadata.
Config summary	~15 min or triggered	Normalized non-secret configuration summary and hash.

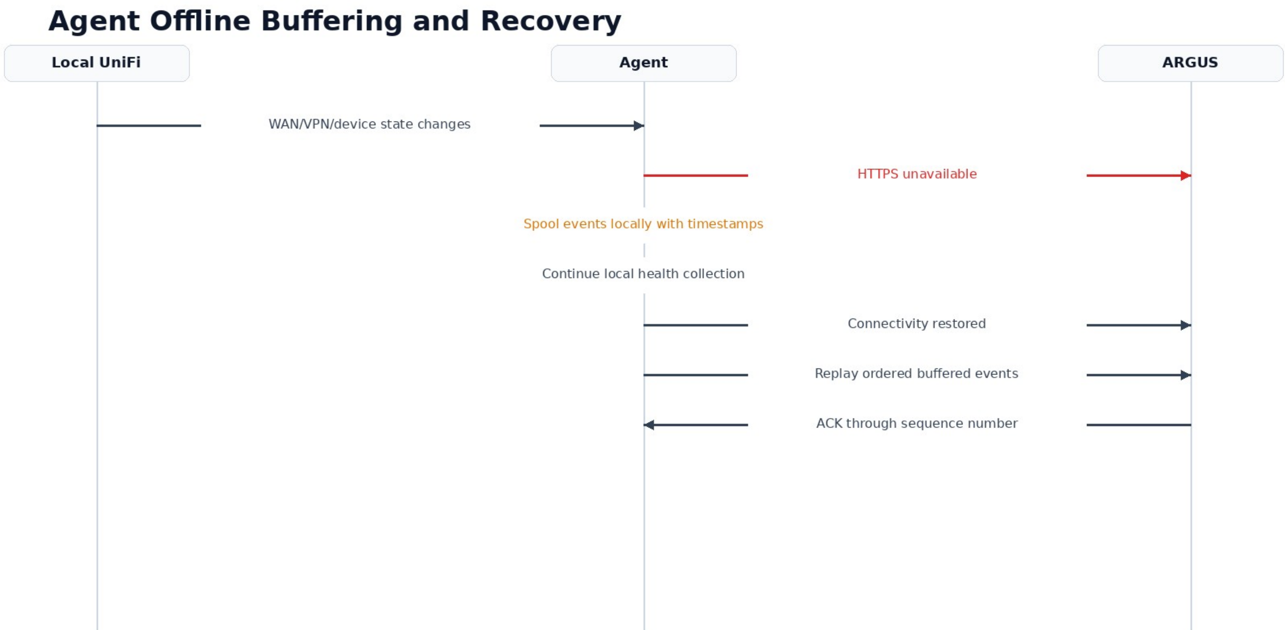


Figure 5.2 - Offline buffering and ordered recovery

6. Enrollment, Identity and Security

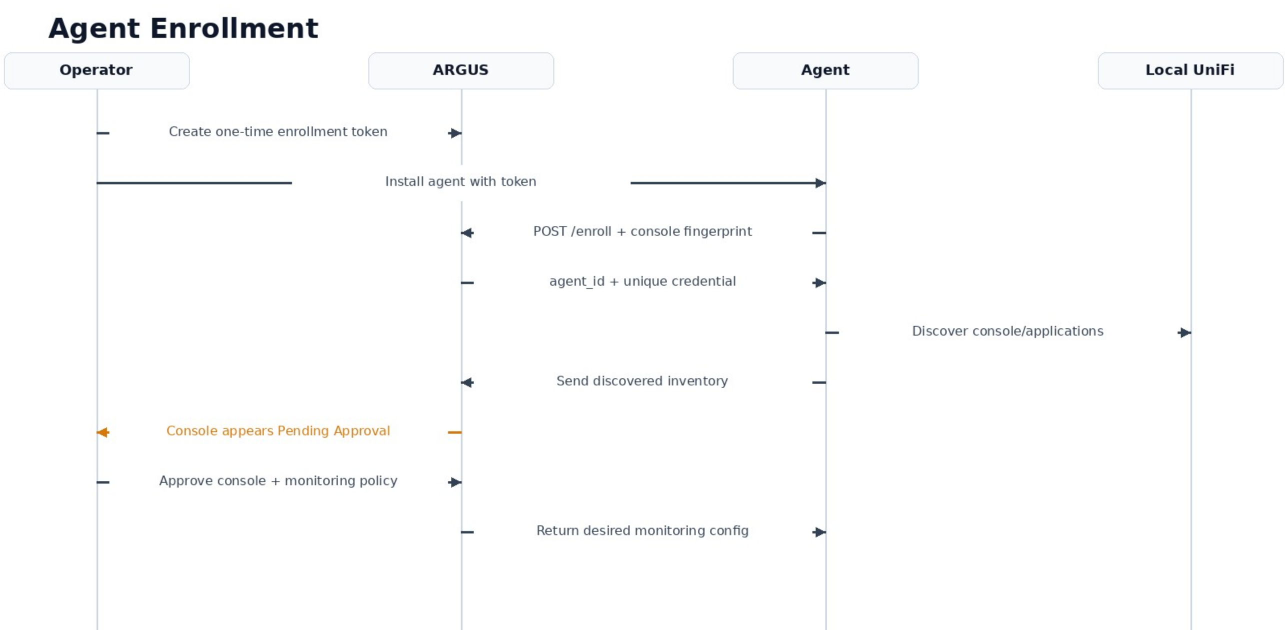


Figure 6.1 - One-time enrollment and approval sequence

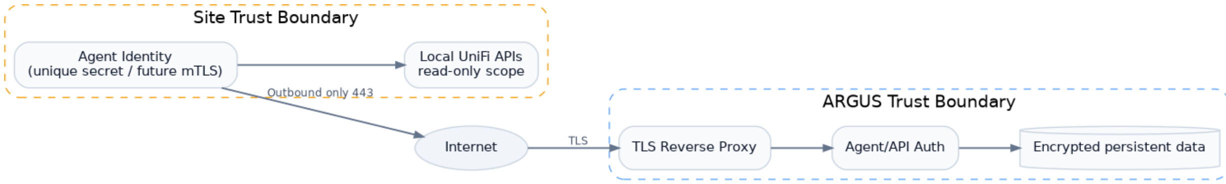


Figure 6.2 - Trust boundaries

6.1 Enrollment model

1. Administrator creates an enrollment token with short TTL and activation count (default: 1 use, 24 hours).
2. MCA-Secure-Downloads-install deploys the Agent and supplies server URL plus token.
3. Agent fingerprints the local console and calls POST /api/v1/enroll over TLS.
4. ARGUS returns an Agent ID and unique long-lived credential; the enrollment token is consumed.
5. Console appears as Pending Approval with discovered inventory preview.
6. Administrator assigns Site and Monitoring Profile, approves the console and optionally selects devices.
7. Agent receives its desired monitoring manifest on the next check-in.

6.2 Security controls

- TLS 1.2+ with modern ciphers; TLS 1.3 preferred.
- Unique credential per Agent, stored with restrictive local permissions.
- Credential rotation and immediate revocation supported.
- Future optional mutual TLS with per-Agent client certificates.
- Rate limits and replay protection using request timestamp/nonce or signed sequence.
- Administrative sessions protected by MFA-capable identity design; local break-glass account considered separately.
- Application-level audit for all security-sensitive actions.
- Secrets excluded from configuration snapshots and logs.

7. Data Acquisition and UniFi Integration

ARGUS should prefer official local UniFi APIs and version-specific Network integration documentation exposed by the local UniFi application. Ubiquiti documents that each UniFi Application has local API endpoints, and the official Network API includes local sites, devices/client-oriented resources, networks, WAN interfaces and site-to-site VPN resources. Site Manager can be used later as an independent secondary source for host/site and ISP metrics, but ARGUS must not depend on cloud reachability for primary site health.

Compatibility rule
Collectors must be capability-driven. During enrollment, the Agent records UniFi OS / Network / Protect versions and enables only the collectors supported by that console. Unsupported fields must degrade to “not available” rather than causing a Site failure.

7.1 Primary vs secondary sources

Source	Role in ARGUS
Local UniFi Network API	Primary source for network sites, networks, devices, WAN and VPN metadata/state when exposed by the installed version.
Local Protect interfaces/API	Primary source for camera/device health when available

	and supported.
Console-local probes	Independent validation for WAN and VPN data-plane health.
UniFi Site Manager API	Optional secondary/correlation source; useful for hosts/sites and ISP metrics, not required for local monitoring.
SNMP	Future fallback/extension for selected metrics or non-UniFi monitoring, not primary in V1.

8. Health Model and State Evaluation

State	Meaning	NOC behavior
HEALTHY	Expected operational state.	Green + “HEALTHY”.
DEGRADED	Functional but abnormal performance/state.	Blue/teal + label; visible but below warnings.
WARNING	Requires review but is not yet a critical outage.	Amber + “WARNING”.
CRITICAL	Confirmed important failure/outage.	Red + “CRITICAL”; highest sort priority.
UNKNOWN	ARGUS cannot determine current condition.	Gray + “UNKNOWN”; never falsely treated as device offline.
DISABLED	Monitoring intentionally disabled or device excluded.	Neutral and excluded from health score.

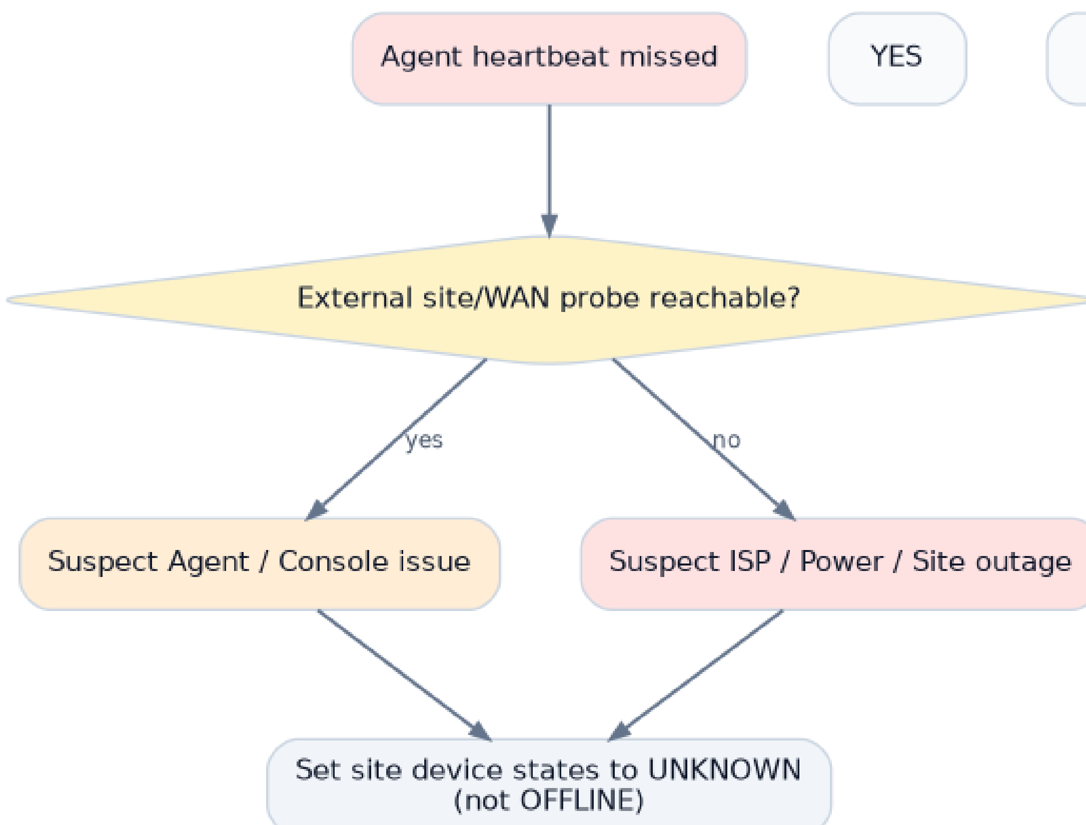


Figure 8.1 - Agent-loss interpretation and UNKNOWN semantics

8.1 Dead-man rules

- Heartbeat target: approximately every 15 seconds with jitter.
- After ~60 seconds without heartbeat: Agent state becomes DEGRADED/WARNING depending on profile.
- After ~90 seconds without heartbeat: Agent becomes CRITICAL and Site dependent health becomes UNKNOWN unless independently known.
- Independent external/WAN probes may classify probable site/ISP/power outage versus Agent/console failure.

9. WAN Monitoring

WAN must be a first-class health component because loss of an uplink can explain Agent/VPN/device behavior. ARGUS should display interface role, operational state, public IP only when policy permits, ISP label, probe latency, packet loss and recent failover state. It should never infer outage from one failed ICMP probe alone; thresholds should require multiple observations or multiple probe targets.

Signal	Purpose
UniFi WAN state	Control-plane view from the console.
External probe from Agent	Confirms data-plane egress.
Latency	Detects significant degradation from baseline/profile threshold.
Packet loss	Detects unstable uplink before full outage.
Failover state	Distinguishes active/standby behavior and WAN transitions.

10. VPN Monitoring

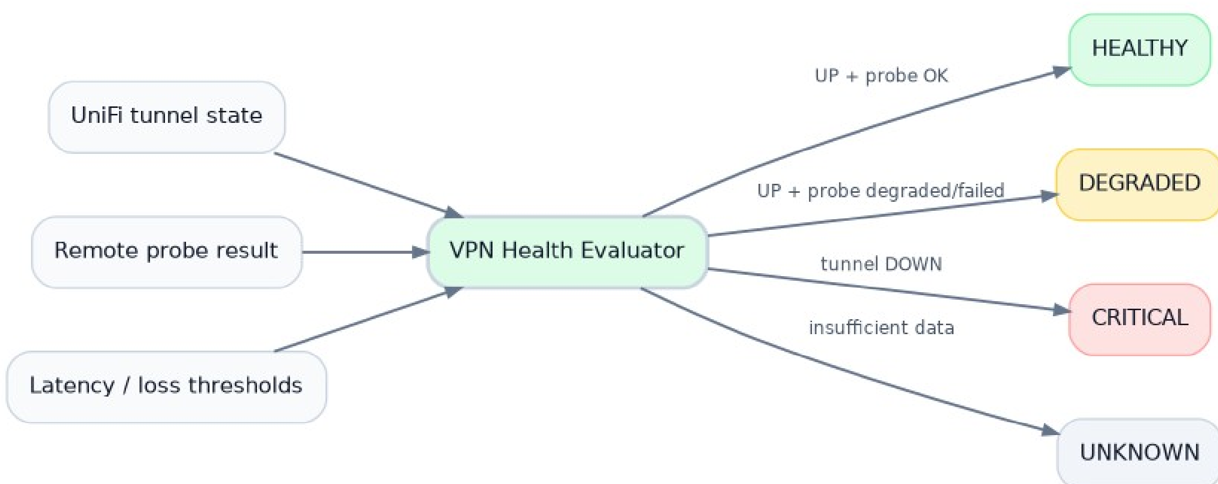


Figure 10.1 - VPN control-plane + data-plane health evaluation

VPN monitoring is explicitly dual-layer. The console's tunnel state is the control-plane signal; an optional remote probe through the tunnel verifies that traffic can actually traverse it. This avoids declaring a tunnel healthy merely because the VPN subsystem reports "connected".

Condition	ARGUS result
Tunnel UP + probe OK	HEALTHY
Tunnel UP + elevated latency/loss	DEGRADED or WARNING by thresholds
Tunnel UP + probe repeatedly failed	WARNING/CRITICAL depending on policy; message identifies data-plane failure

Tunnel DOWN	CRITICAL after debounce/grace
Agent unavailable	UNKNOWN unless an independent source confirms tunnel state
Tunnel administratively disabled	DISABLED

11. Device and Protect Monitoring

Discovery and monitoring are separate states. An Agent may discover every switch, AP and camera, but only approved devices (or those selected by a Monitoring Profile rule) contribute to active health and alerting. This prevents test devices and intentionally disconnected assets from polluting the NOC.

- Store minimal identity: stable UniFi ID, name, model, type, console/site association and monitoring flag.
- Store operational state and timestamps separately from static inventory.
- For Protect, prioritize camera reachable/connected and relevant service/recording indicators over video analytics.
- Do not ingest video, thumbnails or recordings into ARGUS V1.

12. Configuration Summary and Change Detection

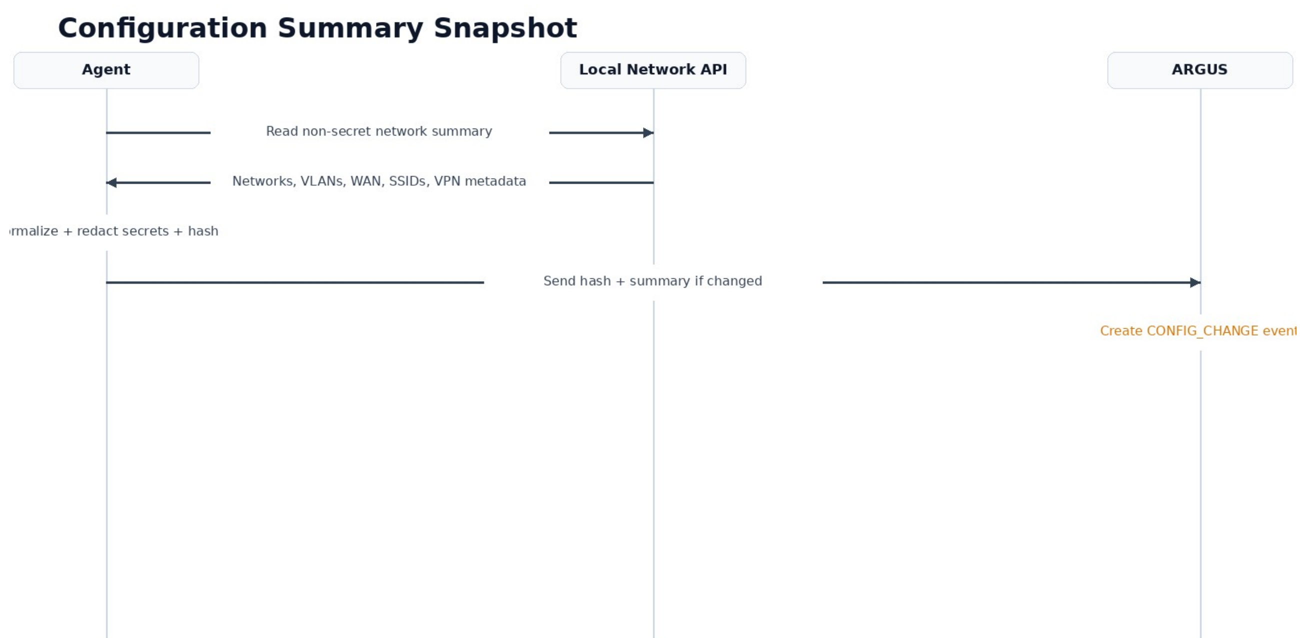


Figure 12.1 - Configuration summary and hash flow

Each Site should expose a concise Network Configuration Summary for support. The Agent normalizes a non-secret subset and calculates a hash. If the hash is unchanged, no duplicate snapshot is written. If it changes, ARGUS stores the new normalized summary and raises a CONFIG_CHANGE event.

Include	Exclude
Network/VLAN names and IDs	WiFi PSKs
Subnets and DHCP enabled/disabled	VPN PSKs
WAN role/state and non-secret addressing metadata	Private keys/cert private material
SSID names and mapped network	RADIUS shared secrets
VPN names, types and remote logical descriptors	Administrative passwords/tokens

High-level firewall-zone/count metadata when supported	Raw config backups in V1

13. Events, Alerts, Incidents and Audit

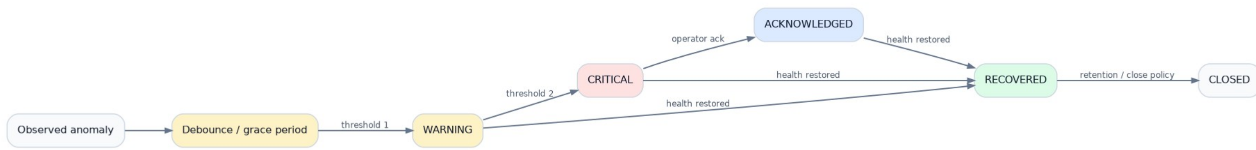


Figure 13.1 - Incident lifecycle

Object	Definition
Event	A timestamped observation or state transition, e.g. VPN_DOWN, DEVICE_RECOVERED, CONFIG_CHANGE.
Alert	A rule outcome requiring operator attention after thresholds/debounce.
Incident	Correlated operational episode with start/end, severity, affected objects and acknowledgement.
Audit	Immutable administrative/security action performed by a user or automation.

Anti-noise rule
Repeated samples that do not change state update “last checked” but do not create new Event rows. An incident remains one incident while repeated failed polls only update its evidence/last_observed timestamp.

14. Data Storage and Retention

Figure 14.1 - Current-state and history retention model

Data class	Suggested retention	Notes
Current state	Indefinite	One latest row per monitored entity/state dimension.
Events	180 days	Configurable; searchable operational history.
Incidents	24 months+	Low volume and high operational value.
Audit	365 days minimum	Longer if compliance requirements demand.
Availability rollups	24 months	Store summarized durations/percentages rather than raw polls.
Agent buffered spool	Size/time bounded locally	For example 24-72 hours, with ordered replay and ACK.

15. Backend/API Architecture

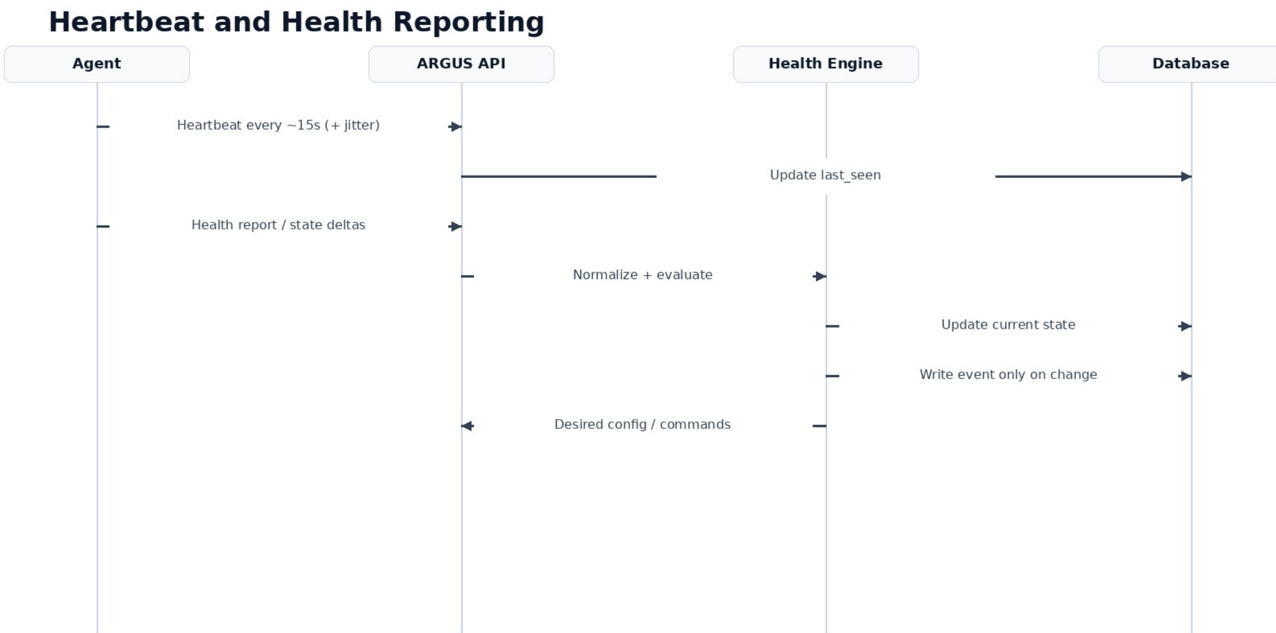


Figure 15.1 - Normal heartbeat and health data flow

All Agent endpoints should be versioned and idempotent where practical. Agent payloads should carry stable entity IDs, source timestamps, sequence numbers and schema version. The server normalizes reports before persistence and health evaluation.

15.1 Conceptual API groups

Path group	Purpose
/api/v1/enroll	One-time Agent enrollment.
/api/v1/agent/heartbeat	Small liveness report.
/api/v1/agent/report	Health/state delta report.
/api/v1/agent/inventory	Discovery/inventory synchronization.
/api/v1/agent/config	Desired monitoring manifest / thresholds / commands.
/api/v1/sites	Administrative site views.
/api/v1/incidents	Incident search, detail and acknowledgements.
/api/v1/events	Filtered event/log retrieval.
/api/v1/admin/*	Users, RBAC, tokens, profiles and settings.

16. AWS Deployment and Operations

- Ubuntu Server 26.04 EC2 instance in a dedicated security group.
- Only required public ingress: TCP/443 (and TCP/80 only if needed for ACME redirect/challenge; DNS-based ACME can avoid it).
- SSH should be restricted through approved administrative controls/VPN/bastion rather than open globally.
- Persistent database volume separated from ephemeral container filesystem.
- Automated encrypted PostgreSQL backups to an AWS-native durable target and periodic restore test.
- CloudWatch or equivalent host-level observability for CPU, disk, memory, instance status and TLS endpoint.

- ARGUS self-health endpoint independent from the main UI where possible.

17. Agent Distribution and Updates

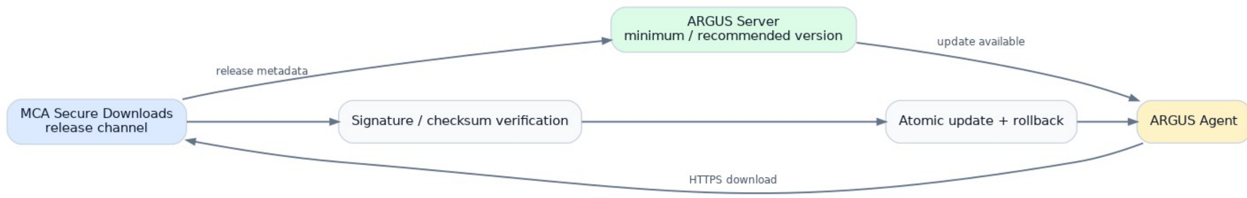


Figure 17.1 - Agent release and update path through MCA Secure Downloads

MCA-Secure-Downloads-install should be the canonical distribution path. Installation must detect supported console/platform, install files with restrictive permissions, register persistence, enroll or accept a supplied token, start the service and verify the first heartbeat. Upgrades should be signed/checksummed, atomic and rollback-capable.

- Stable, optional beta and pinned release channels.
- Server displays installed Agent version per Site.
- Update policy can be manual, staged or automatic by profile.
- No update should destroy the local spool or Agent identity.

18. NOC / 1080p UX Requirements

Hard UI constraint
The primary /noc view must fit entirely inside a browser viewport of 1920×1080 or lower. No vertical or horizontal scroll is allowed in normal wallboard operation.

Requirement	Specification
Viewport	Design baseline 1920×1080; test 1600×900 and 1366×768 fallbacks.
Viewing distance	Primary status readable from approximately 3-5 meters.
Typography	Large KPIs ~40-56 px; site labels ~18-22 px; secondary text >=15-16 px at 1080p.
Status semantics	Color is never the only indicator; pair with text/icon/symbol.
Interaction	No hover-required information; click/tap optional for local operator use.
Sorting	Unhealthy > Unknown > Healthy; longest/most severe incidents first.
Staleness	Persistent “LIVE / last updated” timestamp so frozen displays are obvious.
Focus mode	All / only unhealthy / critical / VPN / WAN / devices.
Rotation	Optional configured sequence of NOC views; pause supported.
Performance	Minimal animation; transitions should not burden Raspberry Pi Chromium.

19. UI Information Architecture

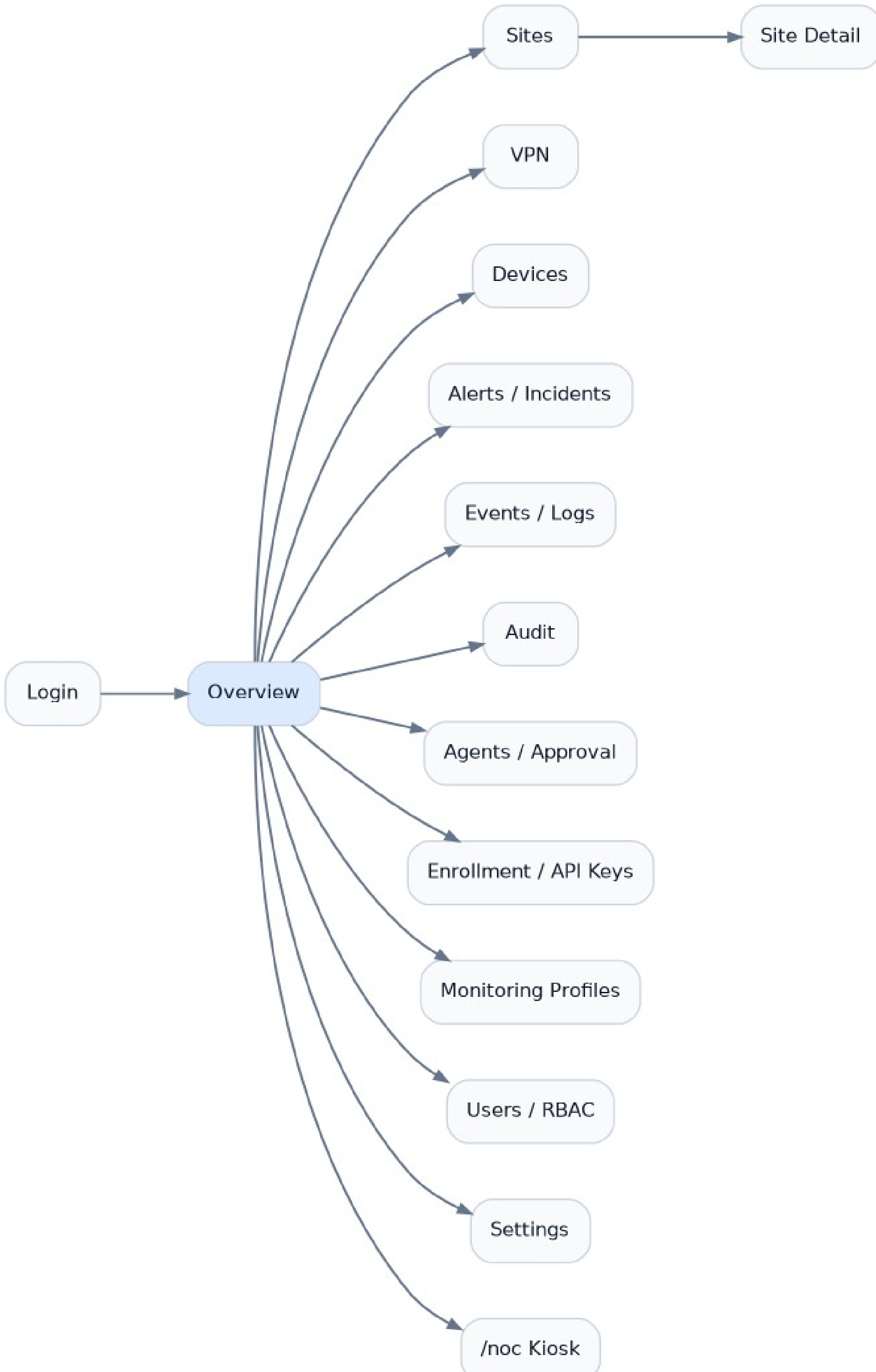


Figure 19.1 - Administrative and NOC navigation map

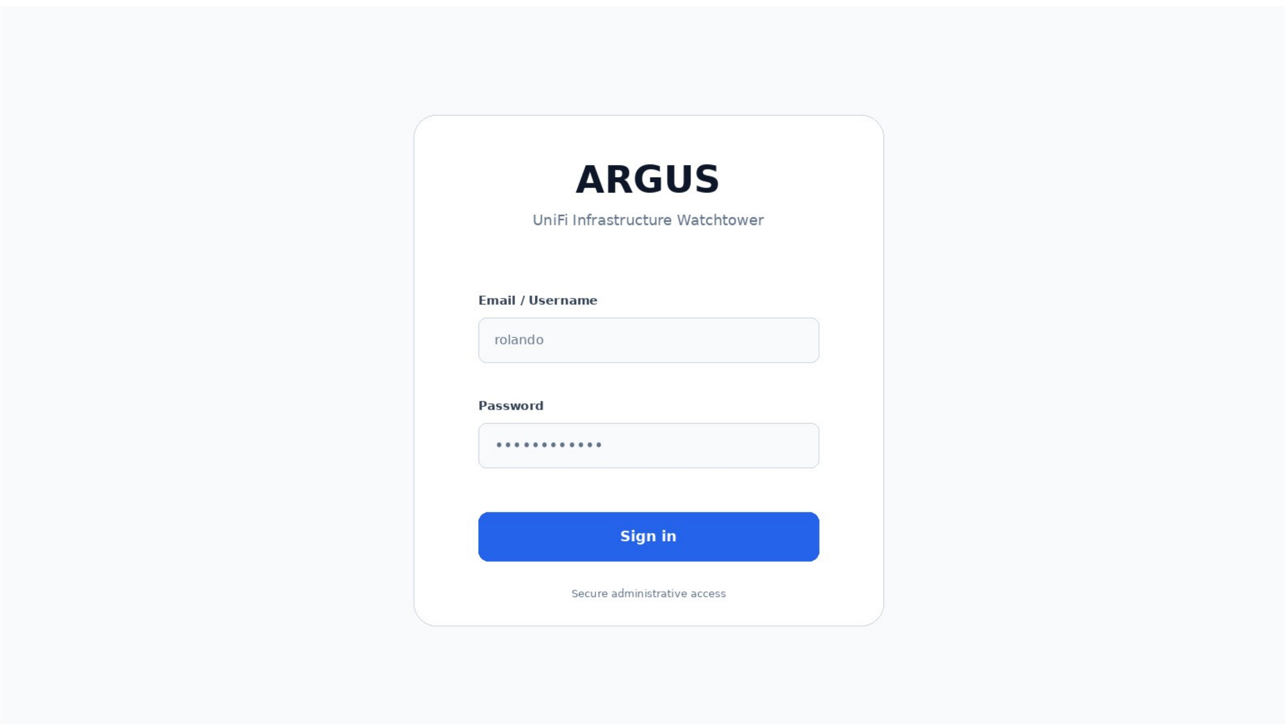
The product intentionally has two presentation modes: the standard administrative application at the root URL and a dedicated /noc wallboard. Both consume the same normalized backend state, but their density, navigation and interaction patterns are different.

20. Storyboards and Screen Specifications

The following wireframes are conceptual but dimensionally consistent 16:9 references. They are not final branding; they establish hierarchy, density, screen ownership and the information that must be visible without scrolling.

20.1 Login

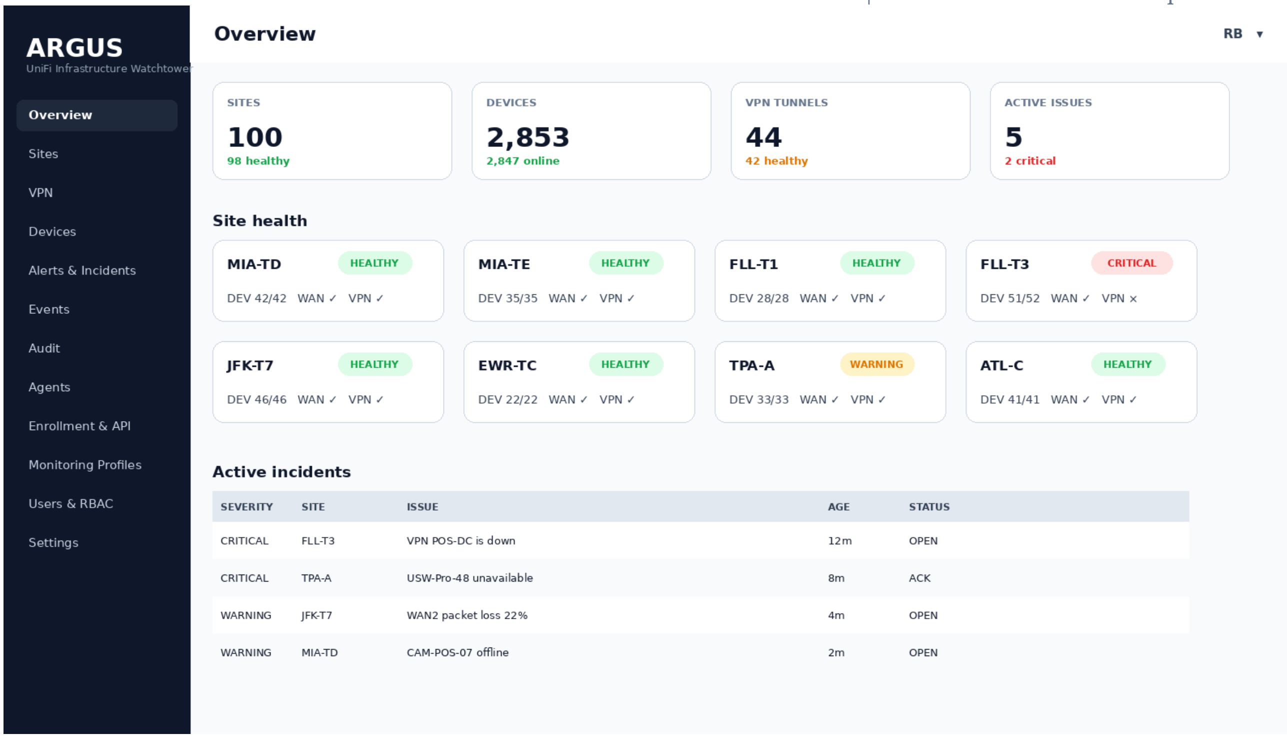
Minimal secure entry point; branding remains restrained so operational UI loads immediately after authentication.



Storyboard 1 - Login

20.2 Administrative overview

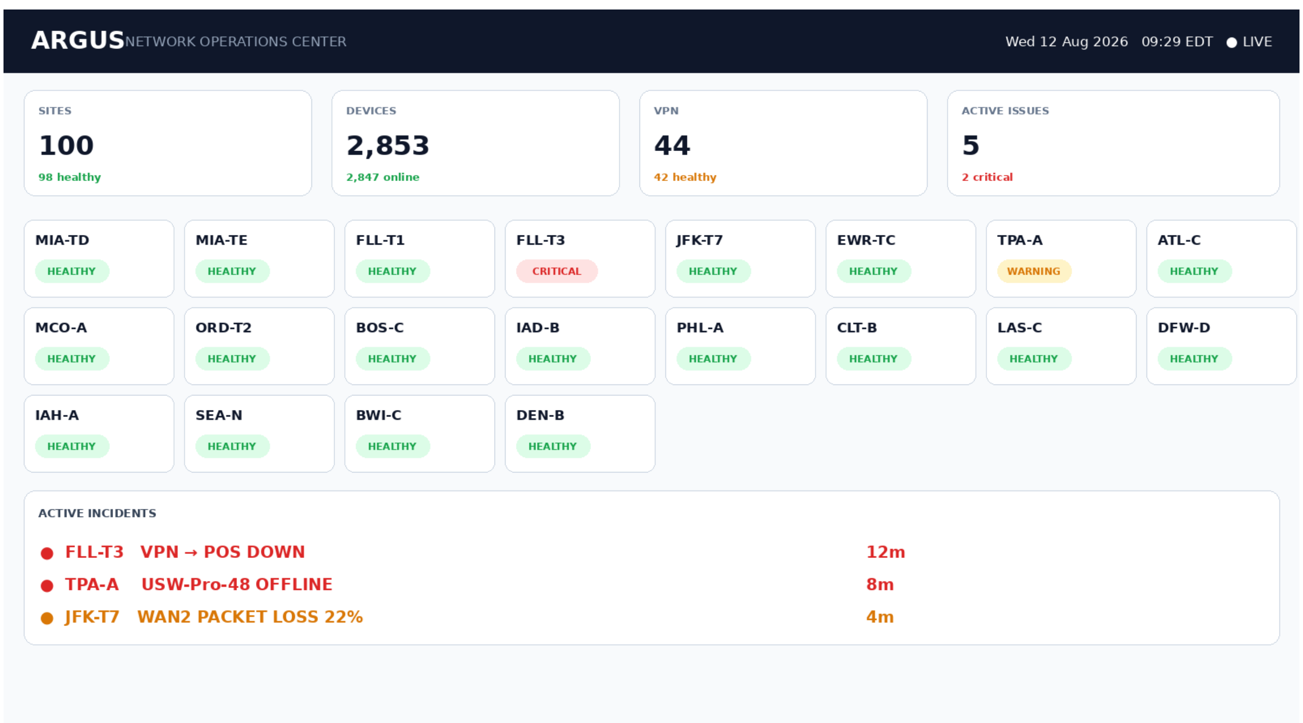
Daily desktop dashboard: compact estate summary, site cards and active incidents.



Storyboard 2 - Administrative overview

20.3 NOC global health

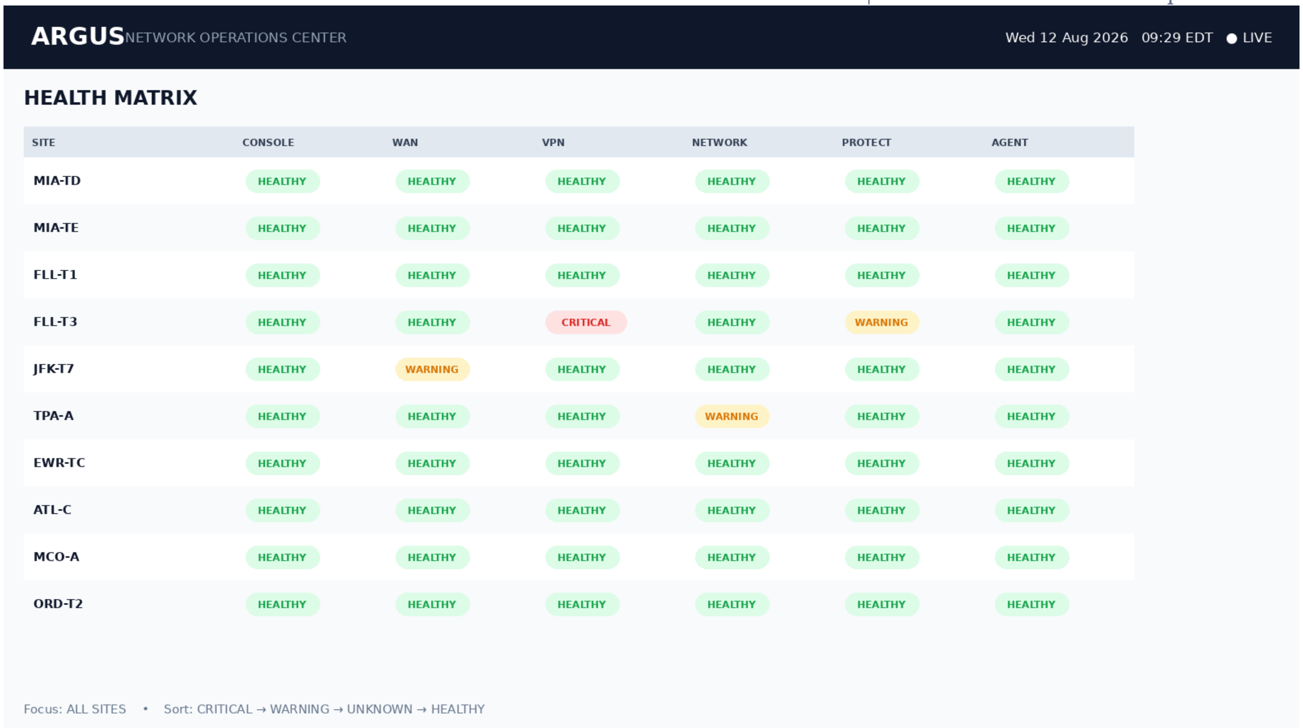
Primary TV wallboard: large KPIs, dense site cards and only the most actionable incidents.



Storyboard 3 - NOC global health

20.4 NOC health matrix

Best at-a-distance diagnostic overview of which health dimension is failing at each site.


ARGUS NETWORK OPERATIONS CENTER
Wed 12 Aug 2026 09:29 EDT ● LIVE

HEALTH MATRIX

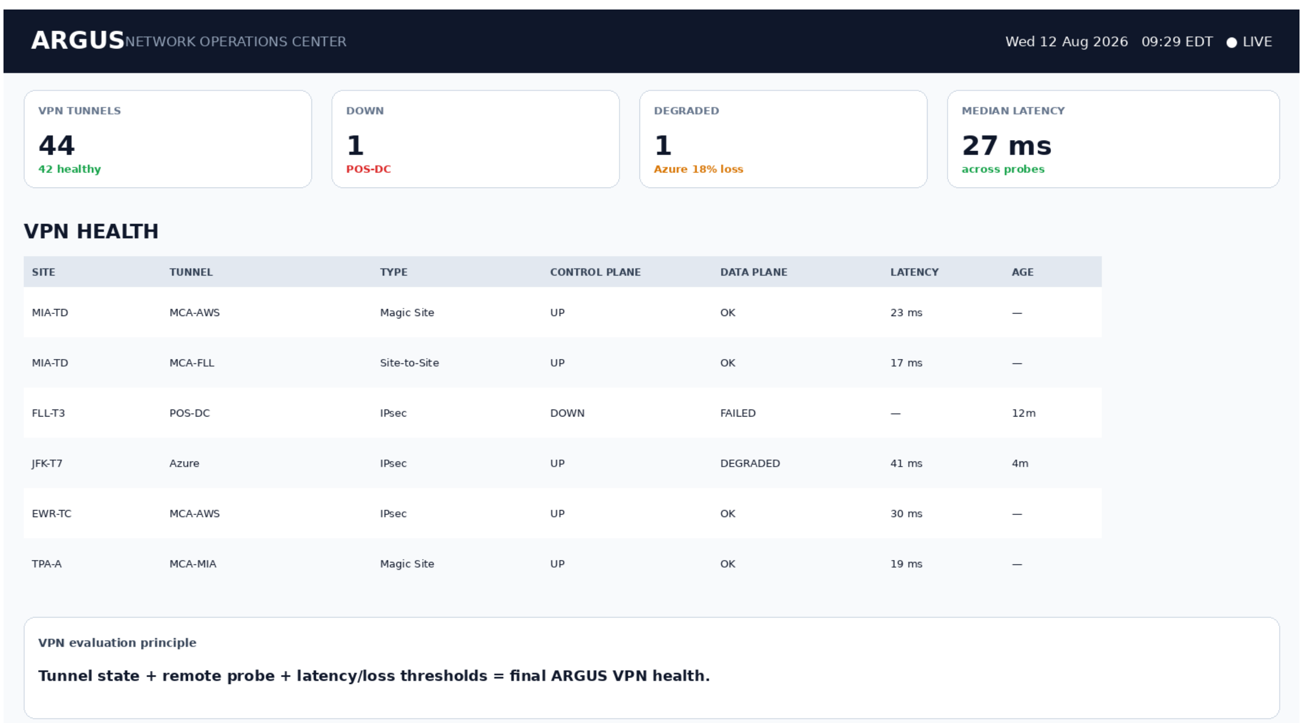
SITE	CONSOLE	WAN	VPN	NETWORK	PROTECT	AGENT
MIA-TD	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY
MIA-TE	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY
FLL-T1	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY
FLL-T3	HEALTHY	HEALTHY	CRITICAL	HEALTHY	WARNING	HEALTHY
JFK-T7	HEALTHY	WARNING	HEALTHY	HEALTHY	HEALTHY	HEALTHY
TPA-A	HEALTHY	HEALTHY	HEALTHY	WARNING	HEALTHY	HEALTHY
EWR-TC	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY
ATL-C	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY
MCO-A	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY
ORD-T2	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY	HEALTHY

Focus: ALL SITES • Sort: CRITICAL → WARNING → UNKNOWN → HEALTHY

Storyboard 4 - NOC health matrix

20.5 NOC VPN health

Dedicated VPN wallboard with control-plane and data-plane columns.


ARGUS NETWORK OPERATIONS CENTER
Wed 12 Aug 2026 09:29 EDT ● LIVE

VPN TUNNELS	DOWN	DEGRADED	MEDIAN LATENCY
44	1	1	27 ms
42 healthy	POS-DC	Azure 18% loss	across probes

VPN HEALTH

SITE	TUNNEL	TYPE	CONTROL PLANE	DATA PLANE	LATENCY	AGE
MIA-TD	MCA-AWS	Magic Site	UP	OK	23 ms	—
MIA-TD	MCA-FLL	Site-to-Site	UP	OK	17 ms	—
FLL-T3	POS-DC	IPsec	DOWN	FAILED	—	12m
JFK-T7	Azure	IPsec	UP	DEGRADED	41 ms	4m
EWR-TC	MCA-AWS	IPsec	UP	OK	30 ms	—
TPA-A	MCA-MIA	Magic Site	UP	OK	19 ms	—

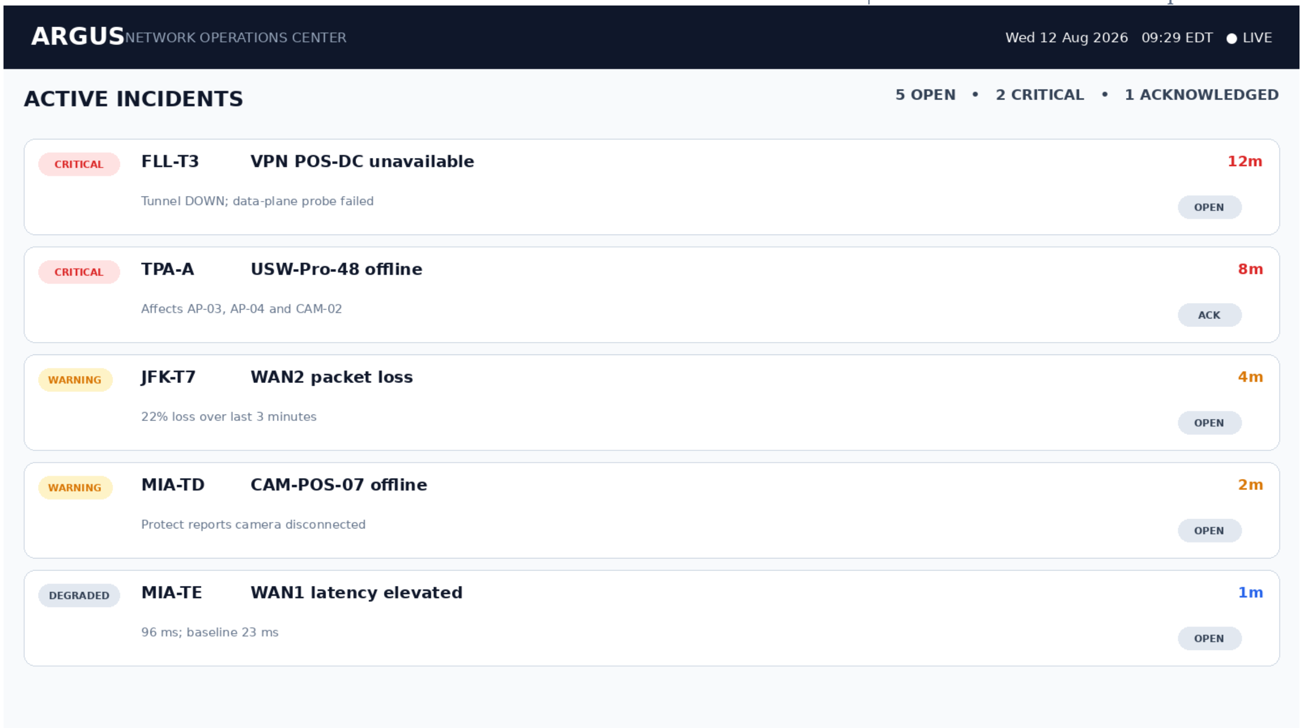
VPN evaluation principle

Tunnel state + remote probe + latency/loss thresholds = final ARGUS VPN health.

Storyboard 5 - NOC VPN health

20.6 NOC active incidents

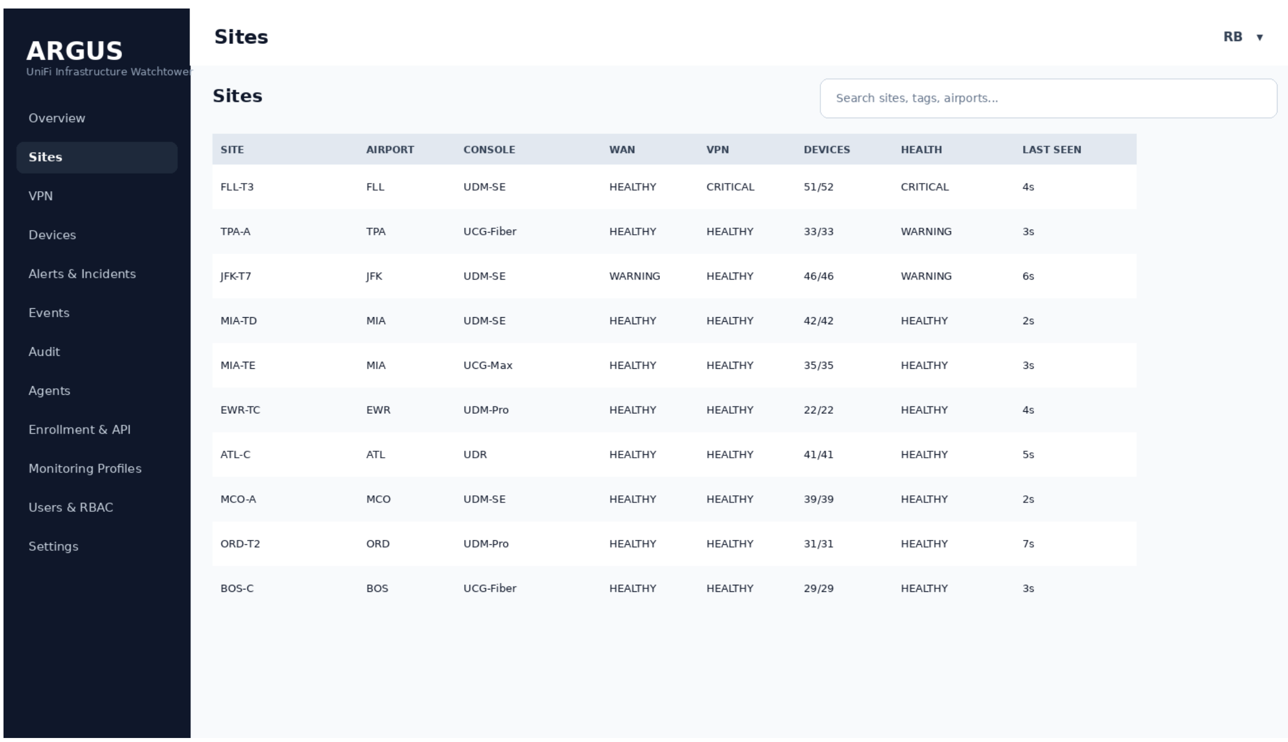
Incident-centric rotation view for active operational work.



Storyboard 6 - NOC active incidents

20.7 Sites list

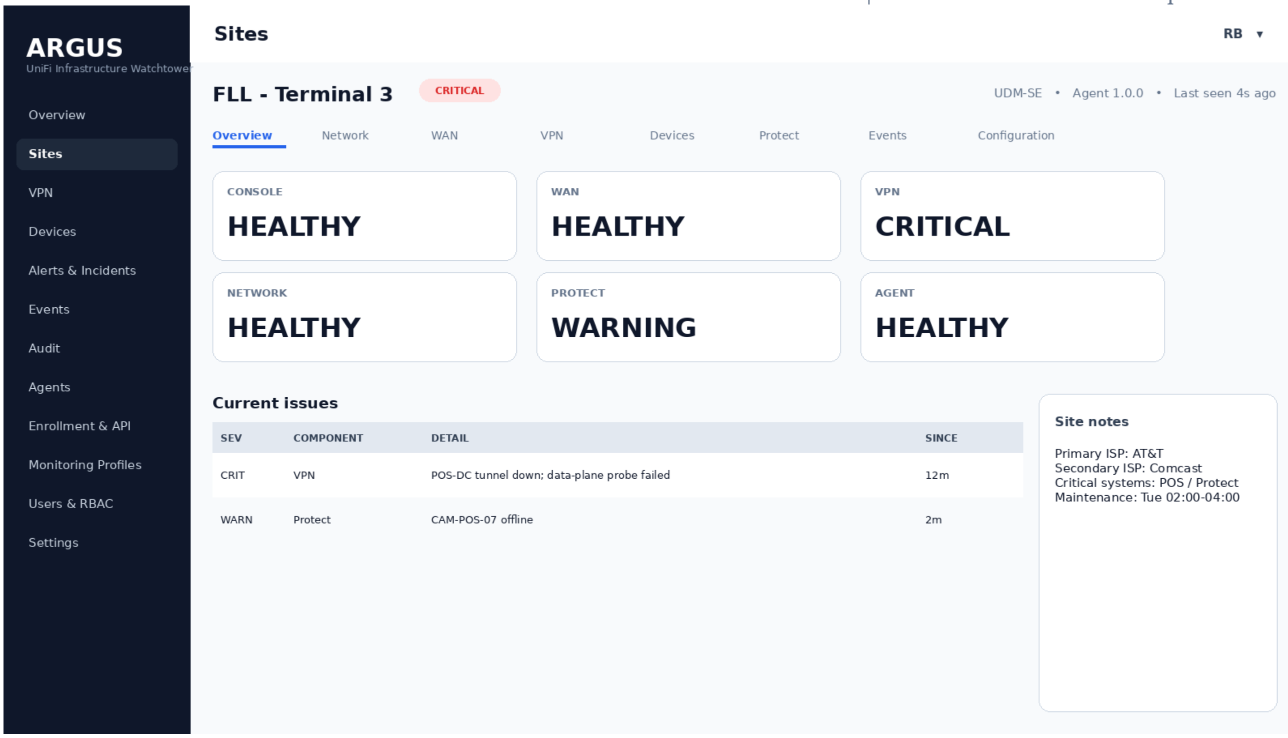
Sortable/filterable estate inventory, unhealthy entries automatically prioritized.



Storyboard 7 - Sites list

20.8 Site overview

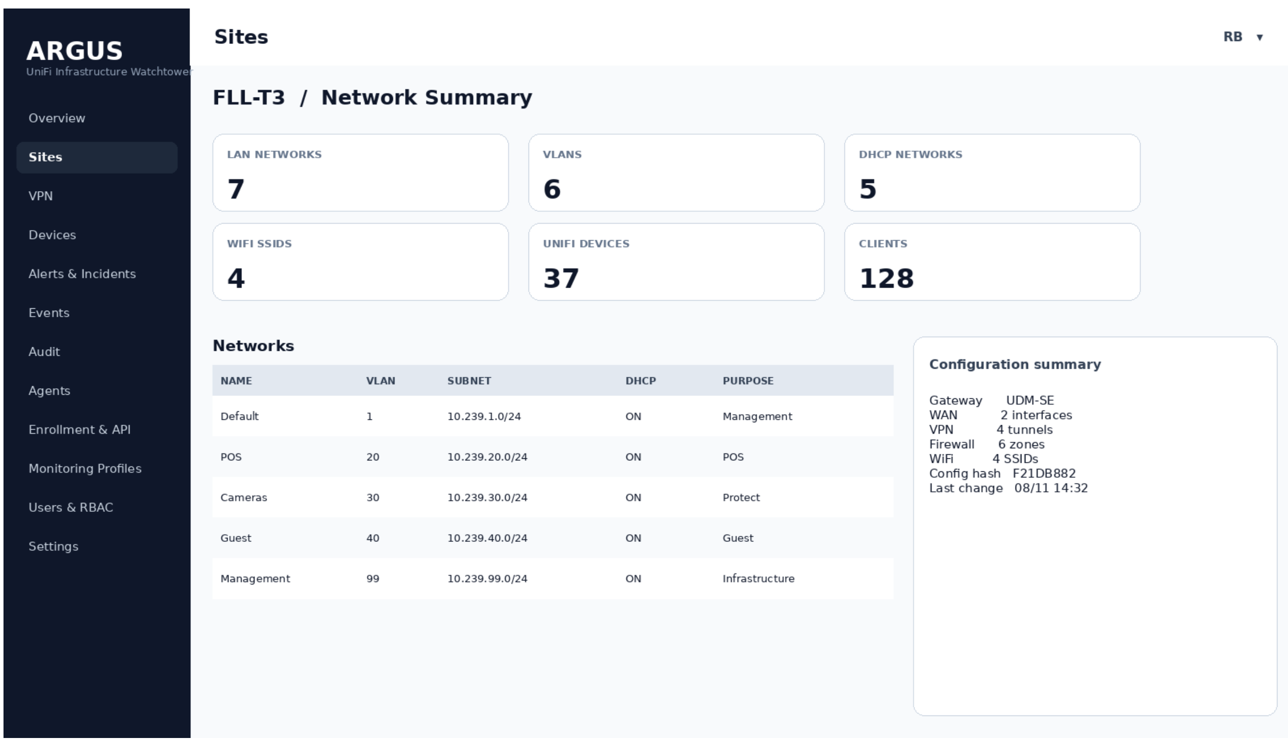
Single-click site synopsis: Console, WAN, VPN, Network, Protect and Agent health.



Storyboard 8 - Site overview

20.9 Site network summary

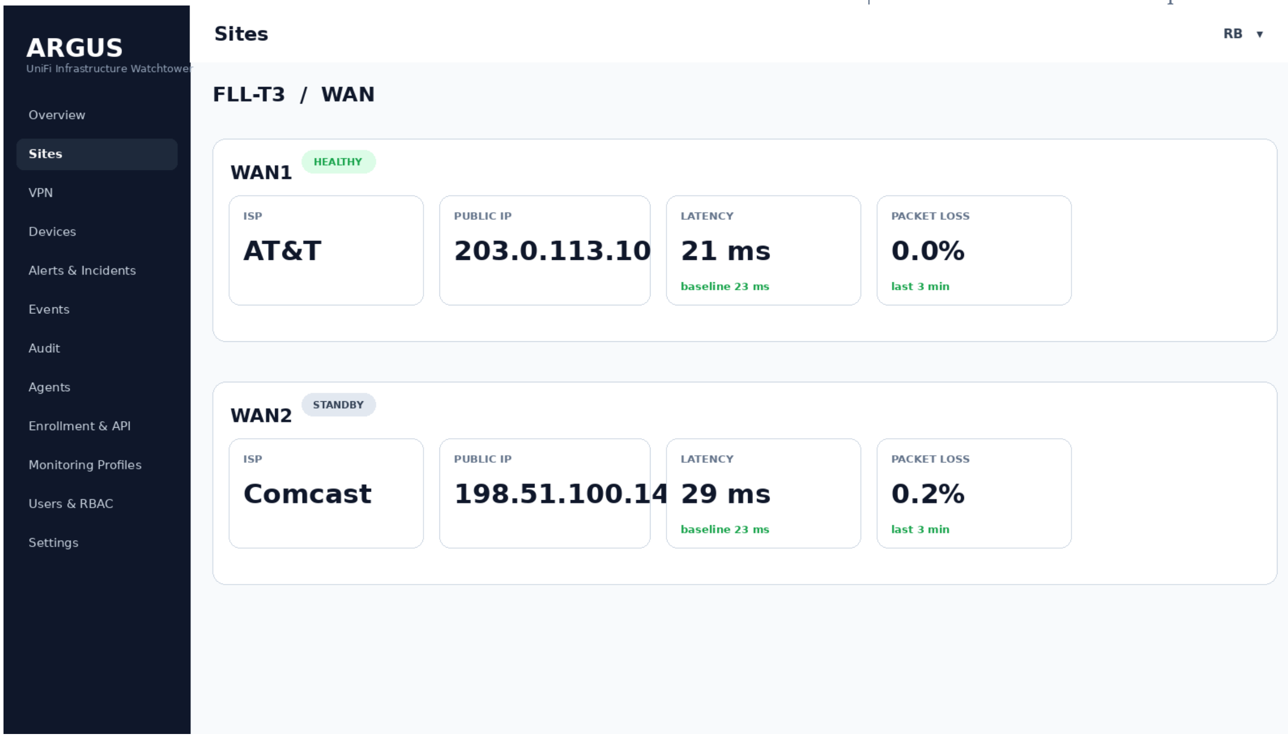
Concise support-oriented network configuration, not a duplicate of UniFi Settings.



Storyboard 9 - Site network summary

20.10 Site WAN

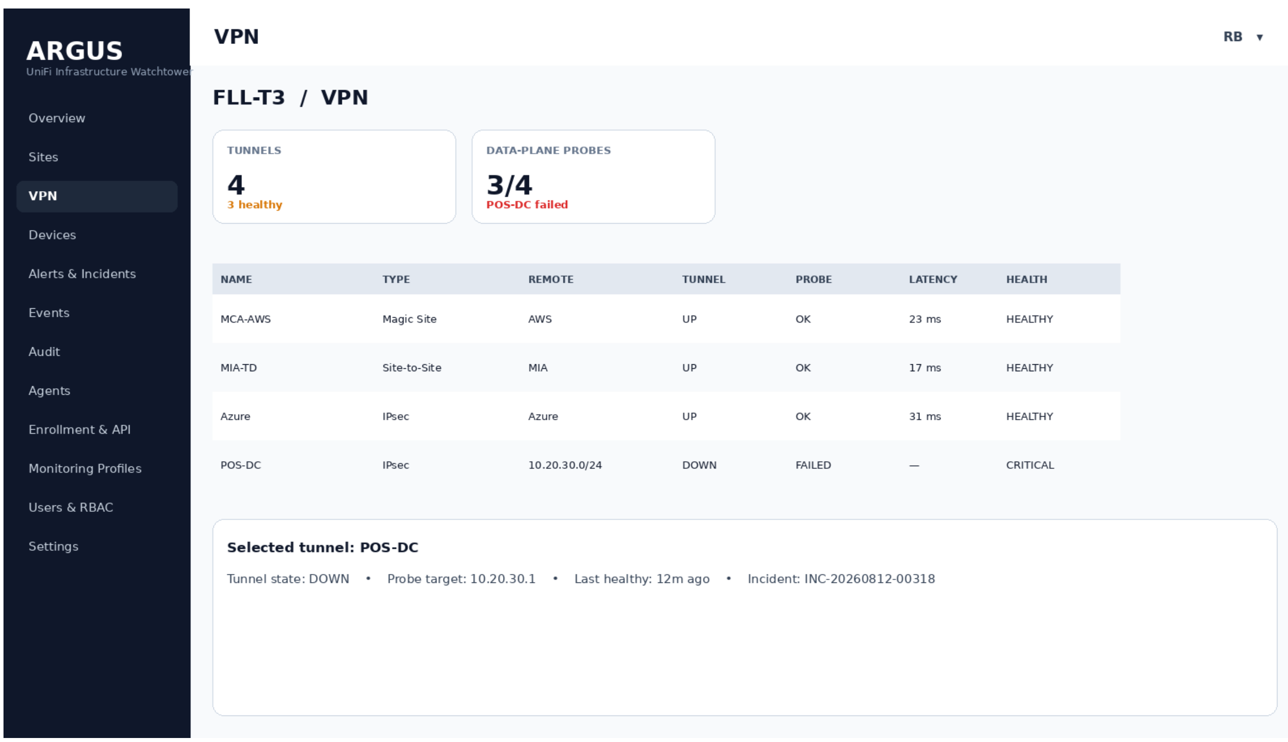
WAN state, ISP, addressing metadata and probe quality.



Storyboard 10 - Site WAN

20.11 Site VPN detail

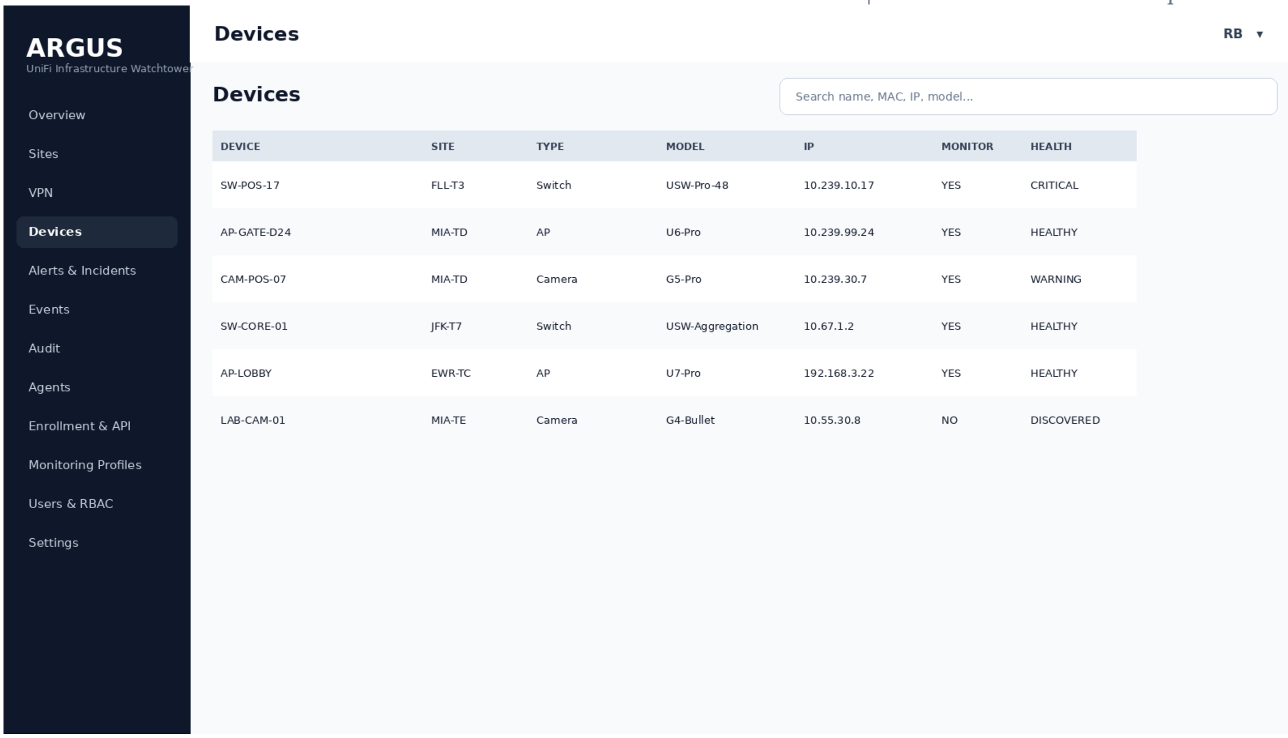
Tunnel inventory and selected-tunnel evidence.



Storyboard 11 - Site VPN detail

20.12 Devices inventory

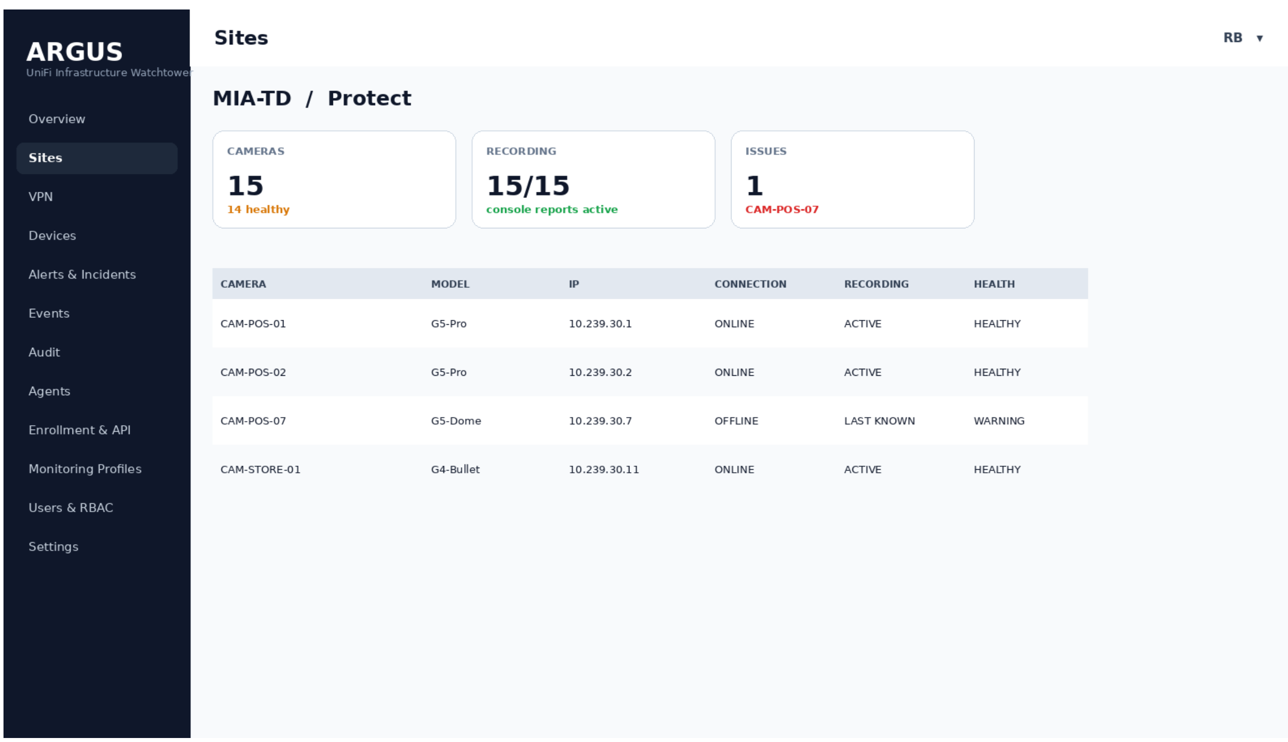
Unified searchable device list with monitoring inclusion state.



Storyboard 12 - Devices inventory

20.13 Protect summary

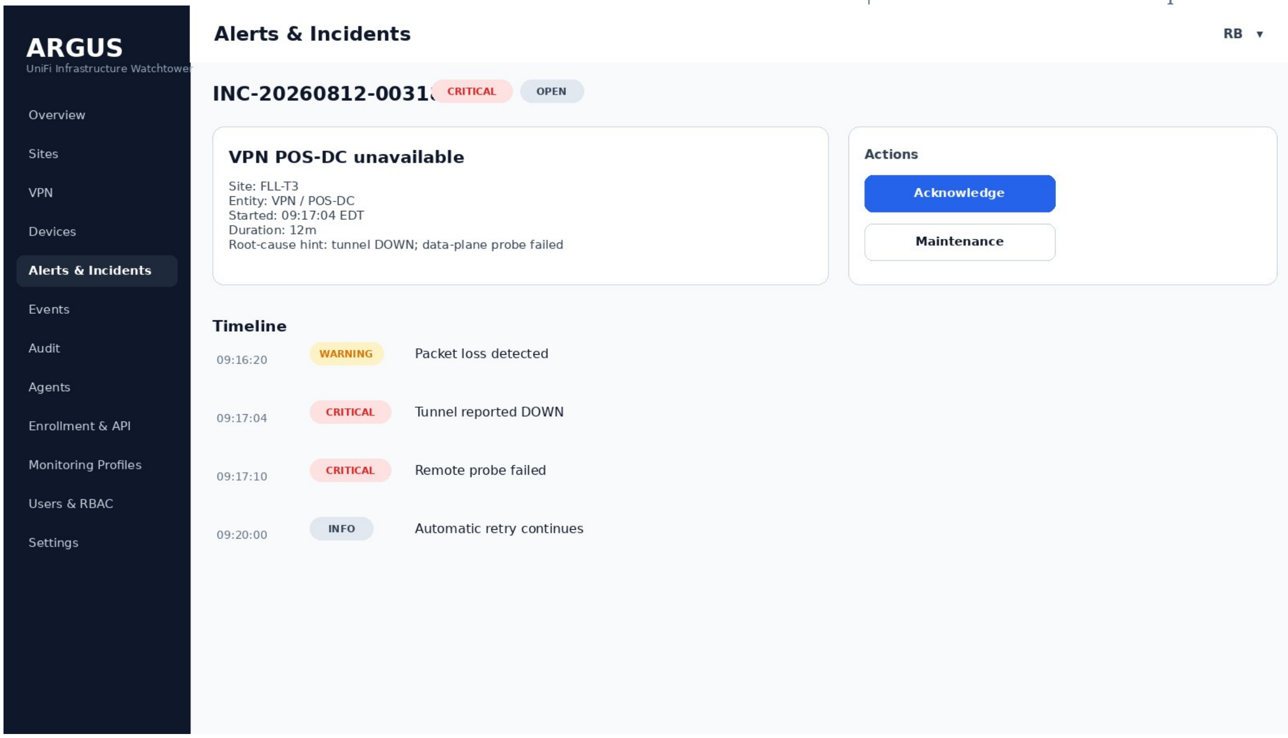
Camera/device operational health only; no video ingestion.



Storyboard 13 - Protect summary

20.14 Incident detail

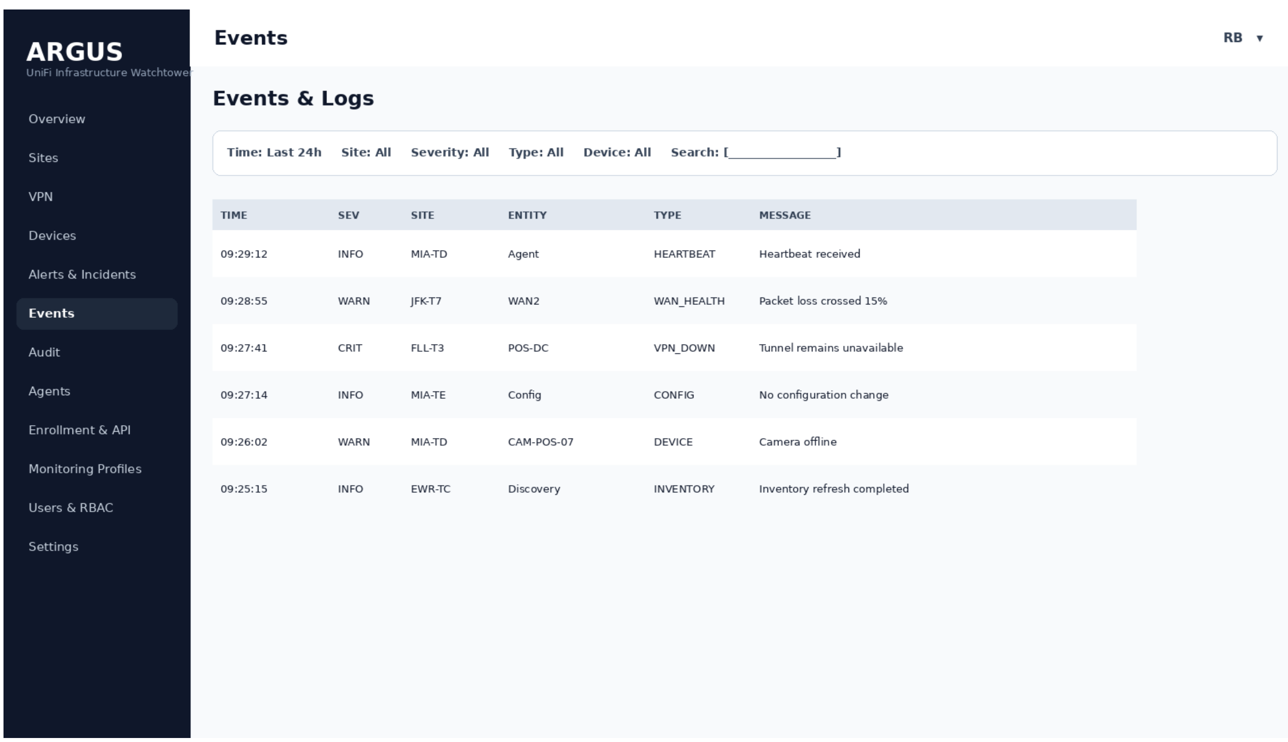
One incident timeline with evidence, acknowledgement and maintenance actions.



Storyboard 14 - Incident detail

20.15 Events and logs

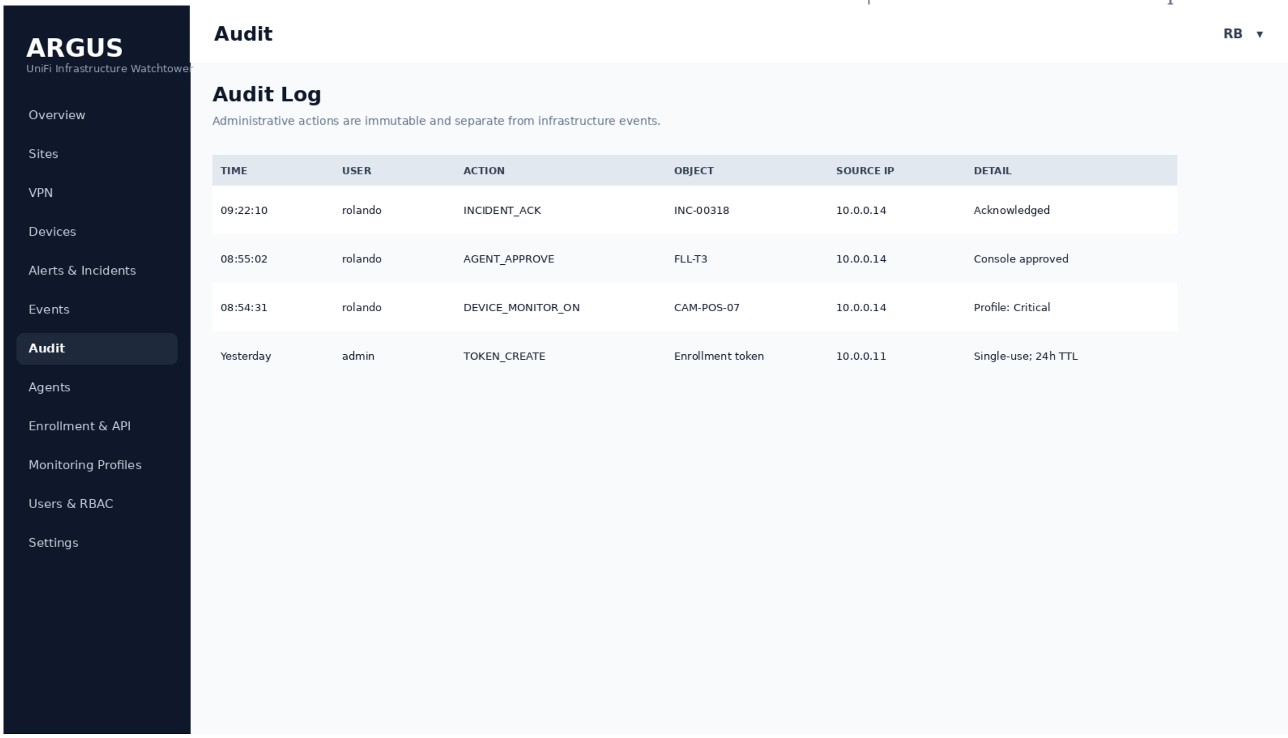
Fast filters and full-text search for operational records.



Storyboard 15 - Events and logs

20.16 Audit log

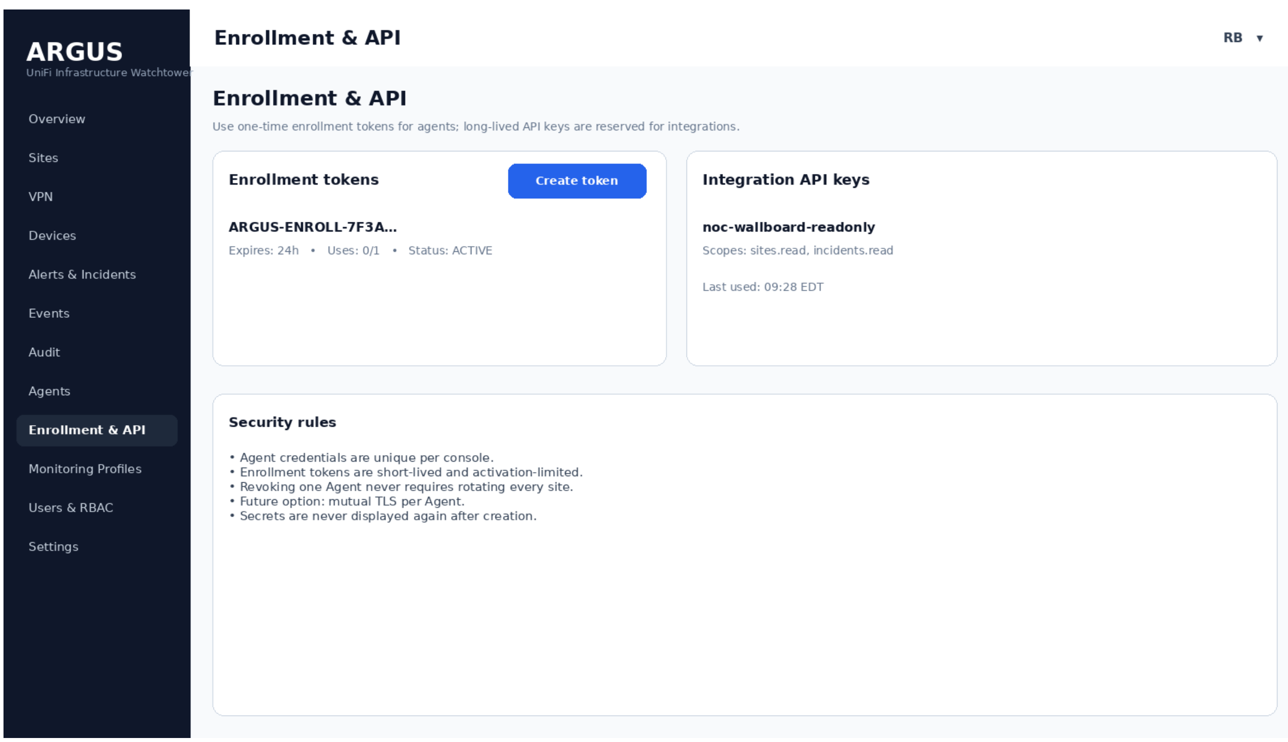
Separate immutable record of administrative actions.



Storyboard 16 - Audit log

20.17 Enrollment and API keys

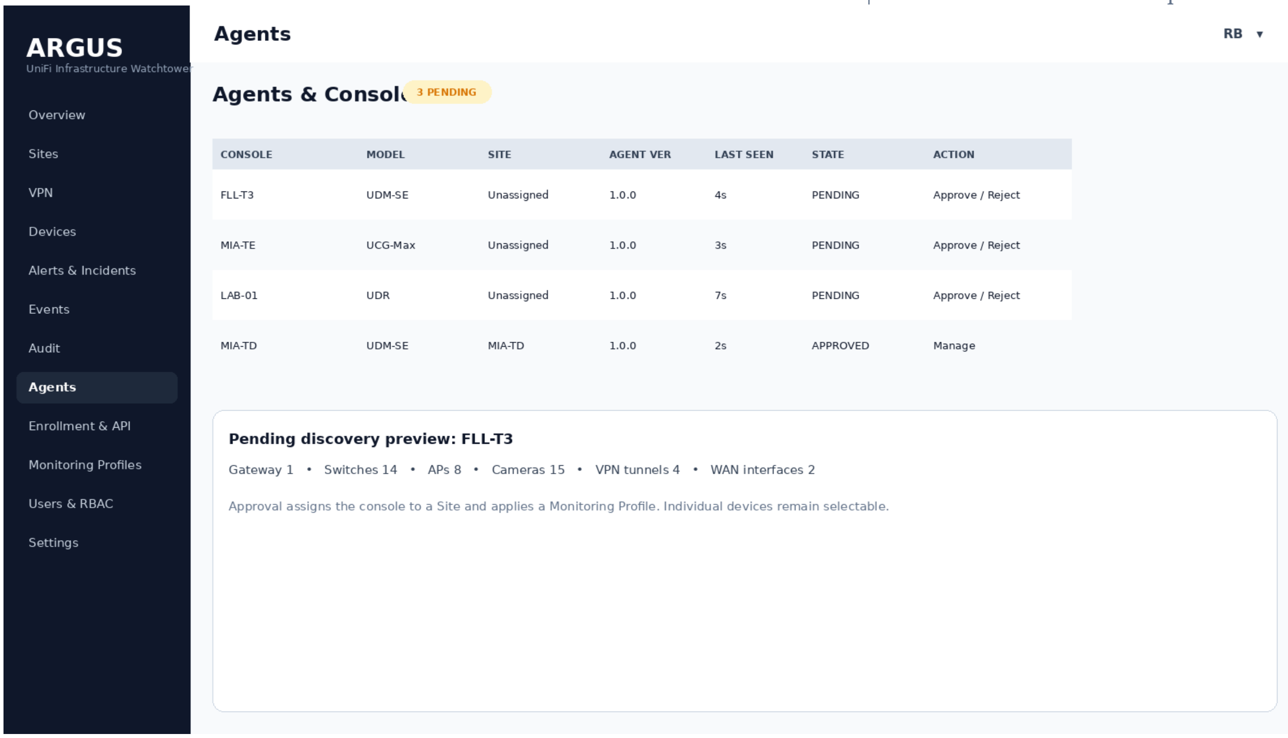
One-time Agent enrollment separate from integration API keys.



Storyboard 17 - Enrollment and API keys

20.18 Agents and approval

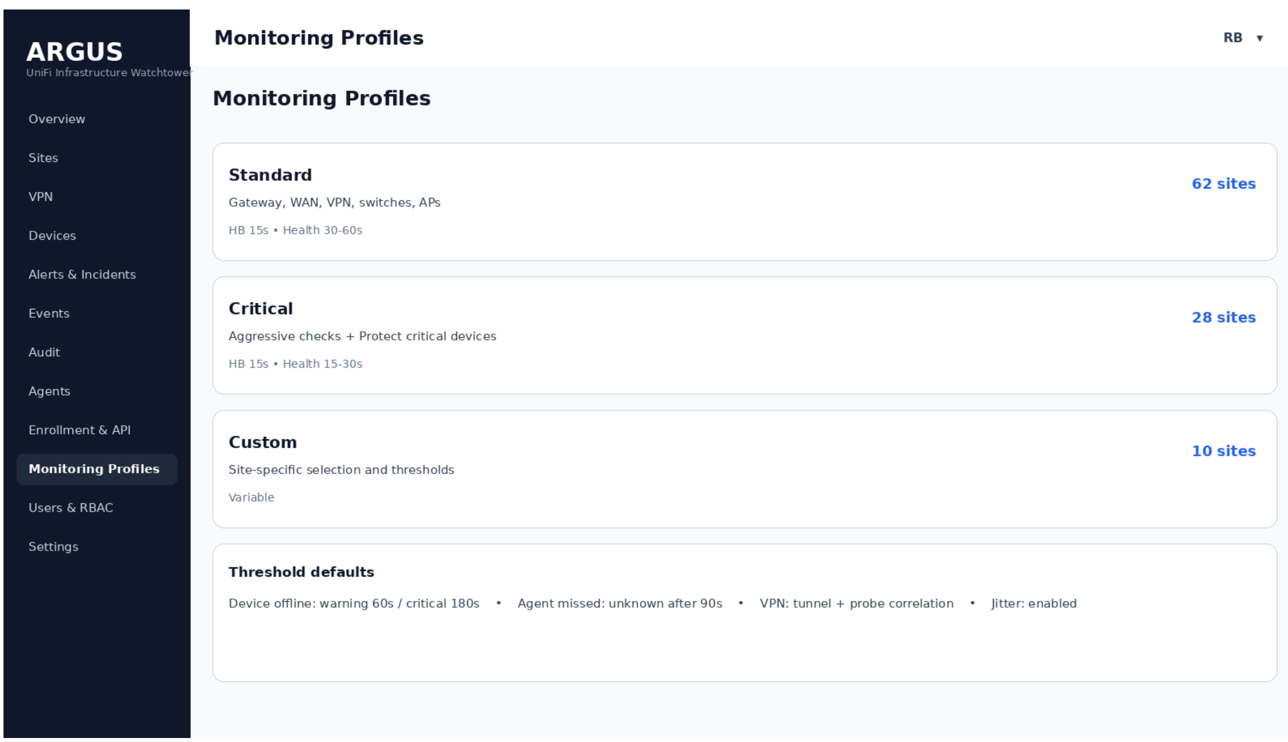
Pending consoles show discovery preview before being admitted to active monitoring.



Storyboard 18 - Agents and approval

20.19 Monitoring profiles

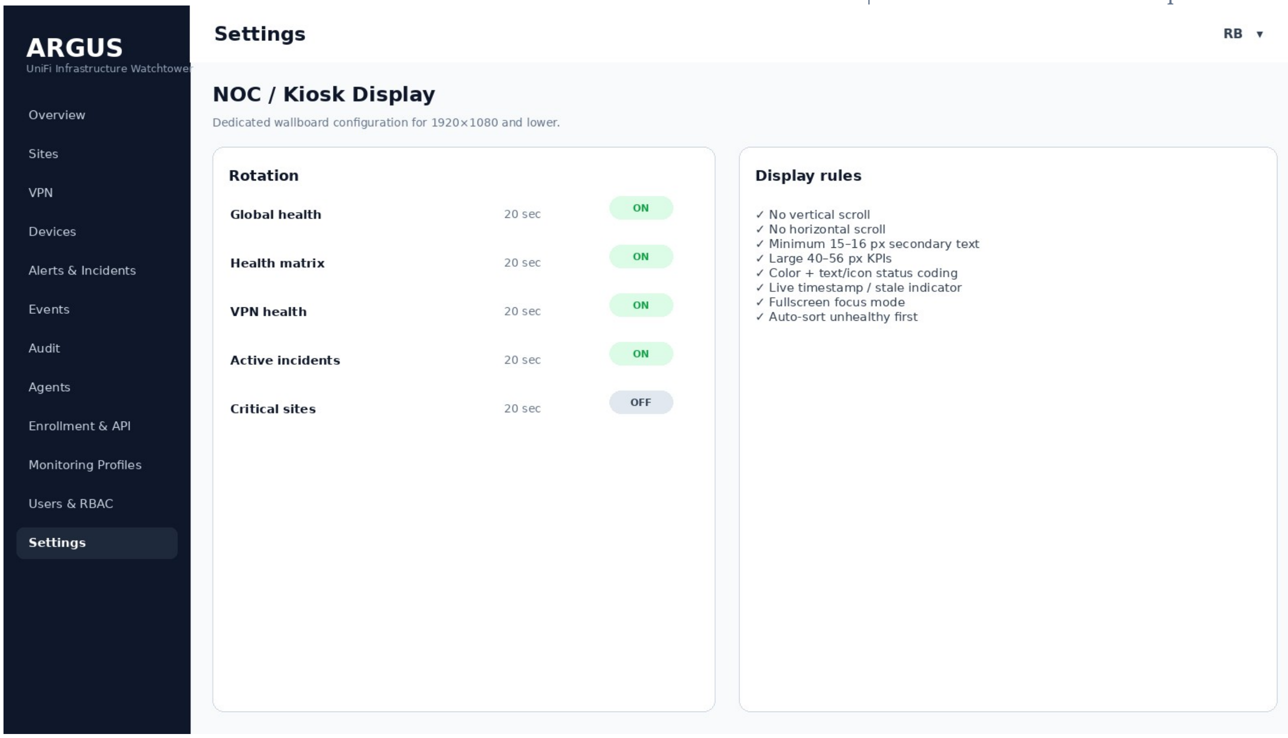
Scale controls for 100+ sites; central cadence/threshold inheritance.



Storyboard 19 - Monitoring profiles

20.20 NOC display settings

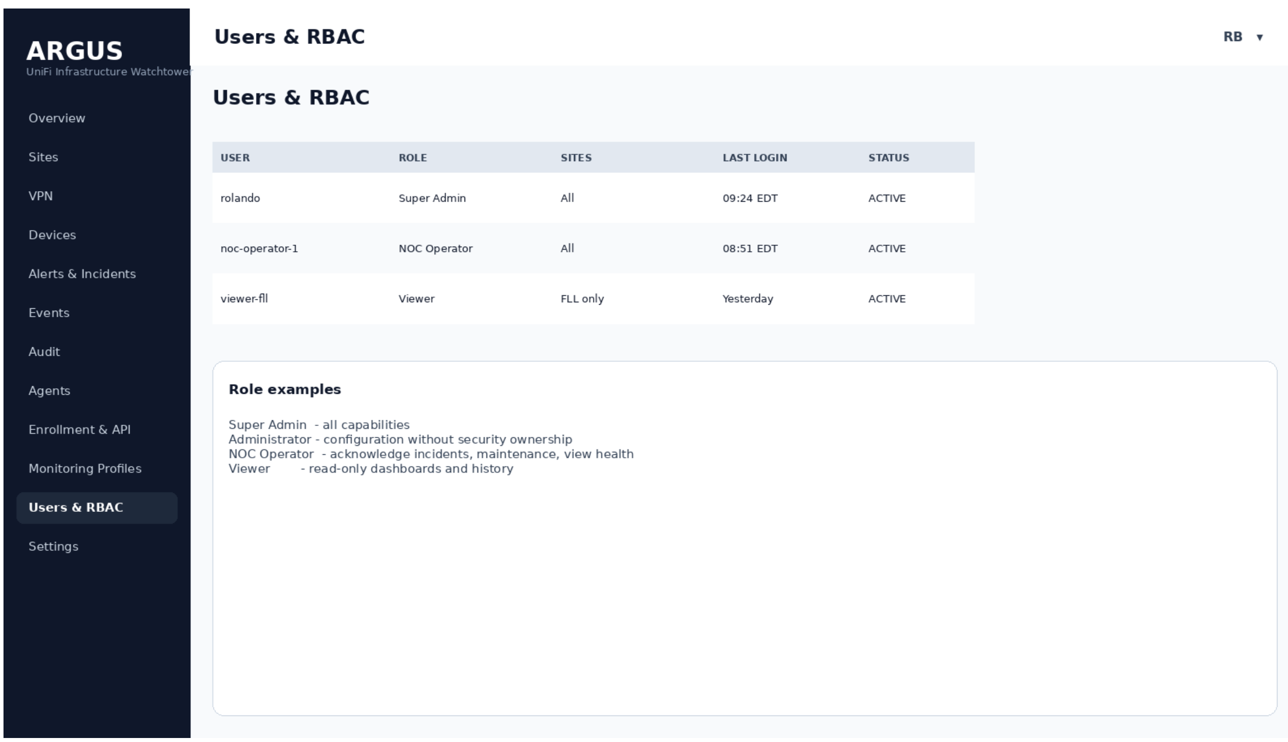
Wallboard rotation and non-scroll display constraints are configurable but safe by default.



Storyboard 20 - NOC display settings

20.21 Users and RBAC

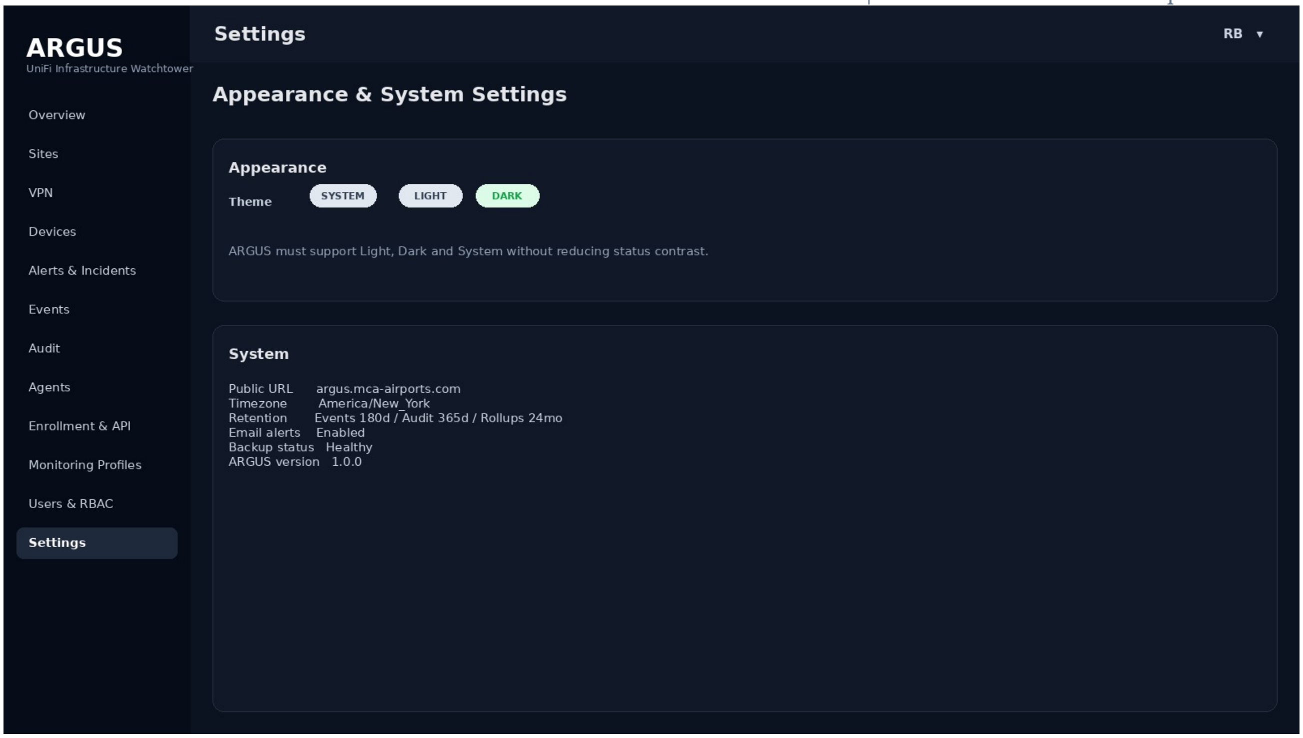
Least-privilege roles for Super Admin, Administrator, NOC Operator and Viewer.



Storyboard 21 - Users and RBAC

20.22 Settings / dark mode

Theme support plus core system/retention configuration.



Storyboard 22 - Settings / dark mode

21. RBAC and Administrative Controls

Role	Typical capabilities
Super Admin	All configuration, security, users, tokens, Agents, Sites, incidents and audit.
Administrator	Operational configuration, Sites, devices, profiles, incidents; optionally restricted security ownership.
NOC Operator	View all health, search logs, acknowledge incidents, set approved maintenance windows, add incident notes.
Viewer	Read-only dashboards, Sites, devices, VPN, incidents and history within authorized scope.

22. Monitoring Profiles and Thresholds

Profile	Default intent	Illustrative cadence
Standard	Normal production sites	Heartbeat 15s; critical health 30s; devices 60s; inventory 5m.
Critical	High-value sites / POS-critical infrastructure	Heartbeat 15s; WAN/VPN/core devices 15-30s; Protect critical 30s.
Custom	Exceptions	Per-site overrides while retaining centrally managed defaults.

All recurring work should include jitter so approximately 100 Agents do not synchronize against the server on minute boundaries. Monitoring intervals are configuration values, not hard-coded constants.

23. Notifications and Maintenance

- Email in V1 with severity-aware templates.

- Microsoft Teams/webhook integration prepared for V2.
- Alert suppression during scheduled maintenance while Events continue to be recorded.
- Recovery notifications tied to Incident closure, not every successful sample.
- Per-profile notification routing and quiet-hour policy in future releases.

24. Resilience and Failure Semantics

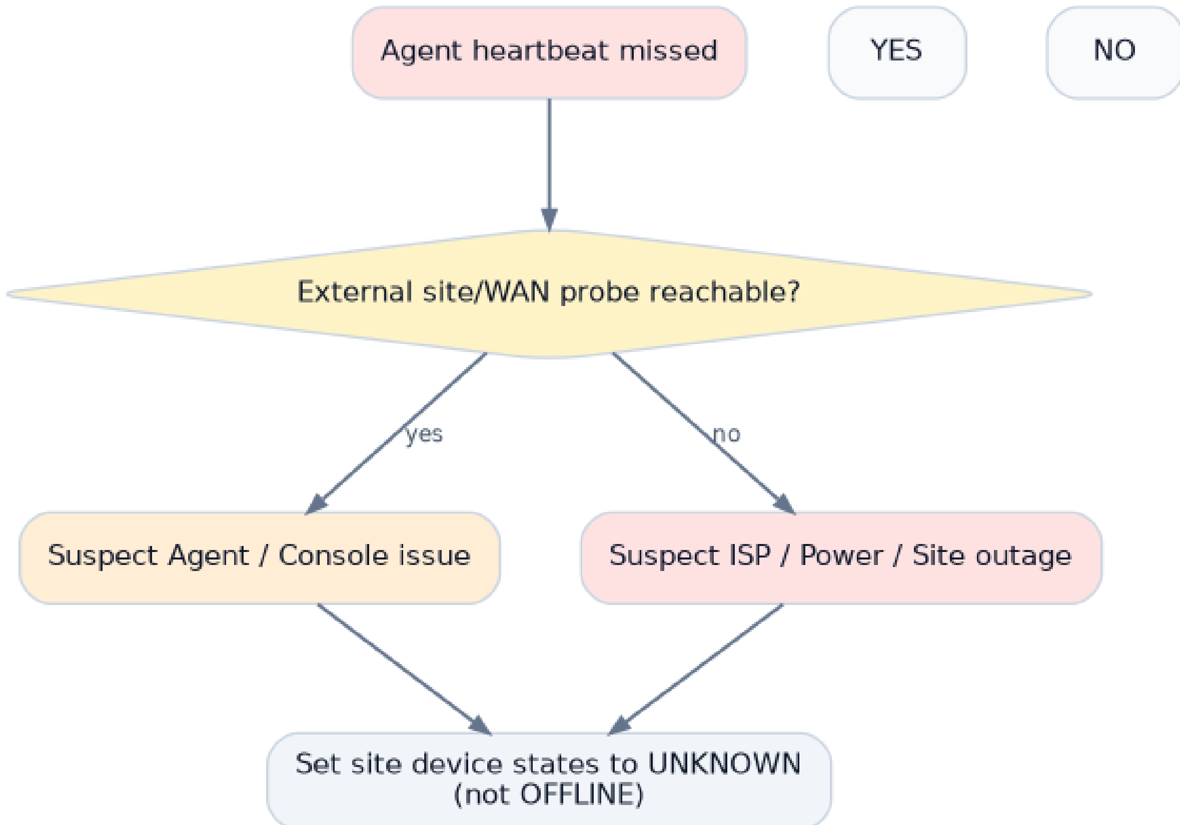


Figure 24.1 - Failure classification on Agent loss

- Agent maintains bounded local spool and continues collection when ARGUS is unreachable.
- Server ACKs replay sequence; Agent deletes acknowledged buffered records safely.
- Server restarts do not require Agent re-enrollment.
- Database outage should produce API backpressure and preserve Agent retry/spool behavior.
- A frozen NOC page must visually expose stale last-update time.
- If ARGUS itself is unhealthy, internal self-monitoring must show API/DB/worker state to administrators and ideally external monitoring must alert independently.

25. Testing and Acceptance Criteria

25.1 Functional acceptance

- Enroll, reject, approve and revoke a console without affecting other Agents.
- Discover representative gateway/switch/AP/camera inventory.
- Select monitored devices and confirm excluded devices do not alert.

- Generate a WAN, VPN, switch/AP and camera failure and observe correct Event → Alert → Incident behavior.
- Demonstrate tunnel-UP/probe-failed VPN degradation.
- Disconnect Agent from Internet, generate local state changes, restore Internet and verify ordered history replay.
- Change a non-secret network property and verify configuration hash/change event.
- Verify maintenance suppresses notifications but keeps history.
- Verify NOC sorting and focus modes.

25.2 UI acceptance

- 1920×1080 /noc loads with no scrollbars.
- 1600×900 remains usable with no clipping.
- 1366×768 fallback remains operational using responsive density rules.
- No critical information depends on hover.
- Red/amber/green statuses always include text/icon equivalents.
- Raspberry Pi Chromium kiosk remains responsive during rotation for an extended soak test.

25.3 Scale acceptance

- Simulate at least 100 connected Agents with jittered heartbeat/report load.
- Verify current-state upserts and event-change suppression prevent runaway table growth.
- Confirm filtered event search and Site overview response times remain interactive under representative retention volume.

26. Rollout Plan

Phase	Sites	Goal
0 - Lab	1-2	Validate local UniFi API collectors, persistence method and enrollment.
1 - Pilot	5	Validate operational accuracy, NOC layouts, VPN probes and offline spool.
2 - Controlled production	20-25	Validate scale, profiles, alert noise, update process and backup/restore.
3 - Fleet rollout	~100	Deploy through MCA-Secure-Downloads-install in staged waves.
4 - Optimization	All	Tune thresholds, retention, dashboards, incident correlation and reporting.

27. Roadmap

- Dependency mapping and root-cause suppression (e.g., switch outage suppresses downstream device storm).
- Site Health Score with weighted criticality.
- Microsoft Teams, webhooks and escalation policies.
- Availability/SLA reporting and historical charts.
- Site Manager ISP metrics correlation.
- Full topology visualization.
- Agent mTLS and automatic certificate rotation.
- Broader SNMP/non-UniFi collectors: UPS, QNAP, Linux, printers, POS health.
- Mobile/PWA operational views.

- Configuration diff details beyond hash-level change detection.

Appendix A - Event Field Baseline

Field	Description
timestamp	Source-observed time in UTC; UI renders configured local timezone.
severity	INFO / DEGRADED / WARNING / CRITICAL.
organization_id	Tenant boundary.
site_id	Operational site.
console_id	Console source.
entity_type	agent / console / wan / vpn / device / protect / config.
entity_id	Stable ARGUS entity ID.
event_type	Machine-readable event name.
message	Human-readable concise explanation.
incident_id	Optional correlated incident.
source_sequence	Agent ordering/replay control when applicable.

Appendix B - Recommended Status Ordering

Default sort order: CRITICAL → WARNING → DEGRADED → UNKNOWN → HEALTHY → DISABLED. Within the same severity, longer-open incidents appear before newer incidents unless an explicit priority override is configured.

Appendix C - Official UniFi API References

The implementation should validate endpoint availability against the locally installed UniFi version during development. Current official documentation referenced for this architecture:

- **Getting Started with the Official UniFi API:** https://help.ui.com/hc/en-us/articles/30076656117655-Getting-Started-with-the-Official-UniFi-API
- **UniFi Network API - Introduction / local application APIs:** https://developer.ui.com/network/
- **Network API - List Local Sites:** https://developer.ui.com/network/v10.4.57/getsiteoverviewpage
- **Network API - List Networks:** https://developer.ui.com/network/v10.0.162/getnetworksoverviewpage
- **Network API - List Site-To-Site VPN Tunnels:** https://developer.ui.com/network/v10.3.58/getsitetositevpntunnelpage
- **Site Manager API - Getting Started:** https://developer.ui.com/site-manager/v1.0.0/gettingstarted
- **Site Manager API - Get ISP Metrics:** https://developer.ui.com/site-manager/v1.0.0/getispmetrics
- **UniFi Local Management:** https://help.ui.com/hc/en-us/articles/28457353760919-UniFi-Local-Management
- **SNMP Monitoring in UniFi Network:** https://help.ui.com/hc/en-us/articles/33502980942615-SNMP-Monitoring-in-UniFi-Network

Appendix D - Diagram Inventory

D01 - System context: D01_system_context.png

D02 - AWS and site deployment topology: D02_deployment_topology.png

D03 - Server component architecture: D03_server_components.png

D04 - Agent component architecture: D04_agent_components.png

D05 - Security trust boundaries: D11_security_boundaries.png

D06 - VPN health evaluation: D10_vpn_health.png

D07 - Health hierarchy: D12_health_hierarchy.png

D08 - Data retention pipeline: D13_retention.png

D09 - Agent update flow: D14_agent_update.png

D10 - UI navigation map: D15_ui_navigation.png

D11 - Outage interpretation decision tree: D16_outage_decision.png

D12 - Incident lifecycle: D09_incident_lifecycle.png

D13 - Core data model: D08_data_model.png

D14 - Enrollment flow: D05_enrollment_sequence.png

D15 - Heartbeat and health flow: D06_health_sequence.png

D16 - Discovery and approval flow: D07_discovery_sequence.png

D17 - Offline buffering and recovery: D17_offline_recovery.png

D18 - Configuration snapshot flow: D18_config_snapshot.png

Appendix E - Storyboard Inventory

S01 - Login: S01_login.png

S02 - Administrative overview: S02_admin_overview.png

S03 - NOC global health: S03_noc_global.png

S04 - NOC health matrix: S04_noc_health_matrix.png

S05 - NOC VPN health: S05_noc_vpn.png

S06 - NOC active incidents: S06_noc_incidents.png

S07 - Sites list: S07_sites_list.png

S08 - Site overview: S08_site_overview.png

S09 - Site network summary: S09_network_summary.png

S10 - Site WAN: S10_site_wan.png

S11 - Site VPN detail: S11_site_vpn.png

S12 - Devices inventory: S12_devices.png

S13 - Protect summary: S13_protect.png

S14 - Incident detail: S14_incident_detail.png

S15 - Events and logs: S15_events.png

S16 - Audit log: S16_audit.png

S17 - Enrollment and API keys: S17_enrollment_keys.png

S18 - Agents and approval: S18_agents_approval.png

S19 - Monitoring profiles: S19_monitor_profiles.png

S20 - NOC display settings: S20_noc_settings.png

S21 - Users and RBAC: S21_rbac.png

S22 - Settings / dark mode: S22_settings_dark.png